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COURSE CODE:

BIO 102

(GENERAL BIOLOGY I)

TAXONOMY + 7 PHY COMPIATION & ECOLOGY

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BIO 102

GENERAL BIOLOGY II

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Department of Animal & Environmental Biology

2018 Lecture Guide

Introduction to Taxonomy

Biologists have identified approximately 1.4 million species of animals, and thousands more are described each year. Many Biologists spend their lives grouping animals according to shared characteristics. These groupings reflect the order found in living systems that is a natural consequence of shared evolutionary processes. Often the work of these Biologists involve describing new species and placing them into their proper relationships with other species.

Early taxonomic system was based on the proposal of Carolus Linnaeus (1753) - which assigns species name organisms based on physical and structural similarities. So, Biologists group animals according to their evolutionary relationships as revealed by ordered patterns in their sharing of homologous features and this called the "natural system" of classification.

Generally, **systematics** is the study of the identification, taxonomy and nomenclature of organisms, including classification of living things with regard to their natural relationships and the study of variation and evolution of taxa. **Taxonomy** is the classification of organisms in an ordered system that indicates natural evolutionary relationships. These studies result in the description of new species and the organization of animals into groups (taxa) based on degree of evolutionary relatedness.

Animal systematics has three major goals:

- (1) to identify all species of animals,
- (2) to evaluate evolutionary relationships among animal species, and
- (3) to group/classify animal species in a hierarchy of taxonomic groups (taxa; singular = taxon) that reflect evolutionary relationships.
- (4) naming

Modern Taxonomy is now based on evolutionary relationships. Taxonomists study:

- ✓ Structural similarities
- ✓ Chromosomal structure (karyotypes)
- ✓ Reproductive potential
- ✓ Biochemical similarities
- ✓ Comparing DNA and amino acids
- ✓ Embryology/development
- ✓ Breeding behavior
- ✓ Geographic distribution

Taxa are ranked to denote increasing inclusiveness as follows: ^{Order of Classification SEFOCPK} species, genus, family, order, class, phylum, and kingdom. All of these ranks can be subdivided to signify taxa that are intermediate between them.

Nomenclature

One of the characteristics of modern humans is our ability to communicate with a spoken language. Language not only allows us to communicate, but also helps us encode and classify concepts, objects, and organisms that we encounter. Common names are sometimes used to refer to organisms, but common names are not officially defined. Also, many animals in different species may share the same common name. Common names are generally not capitalized (except as required grammatically). Examples: tiger shark, timber wolf, daisy, puffball fungus.

However, to make sense out of life's diversity, we need more than just names for organisms. To be useful, a naming system must reflect the order and relationships that arise from evolutionary processes. **Nomenclature** is the assignment of a distinctive name to each species. The current system uses Binomial nomenclature where species of organisms are given 2 names (scientific name).

Thus, names of species are binomial, with the first name designating the **Genus** to which the species belongs followed by a species epithet. The genus, which is a group of similar species, begins with a capital letter, while the species designation begins with a lowercase letter, and the entire scientific name is italicized or underlined because it is derived from Latin. Thus, the scientific name of humans is written *Homo sapiens*. The genus name may be abbreviated to its first letter after it has been printed in full earlier in the same document. e.g. *H. sapiens*. Taxa at all other ranks are given single non-italicized names. However, if another species name with a genus name that starts with "h" is used in the same document, both genus names must be written in full wherever they occur. For example, if *Hirudo medicinalis* is mentioned together with *Homo sapiens*, neither *Homo* nor *Hirudo* should be abbreviated when they occur again in the same document.

The binomial system of nomenclature brings order to a chaotic world of common names. Common names have two problems. First, they vary from country to country, and from region to region within a country. Some species have literally hundreds of different common names. Biology transcends regional and nationalistic boundaries and so must the names of what biologists study. A common name, even if you recognize it, often does not specify a particular species. The binomial system of nomenclature is universal and clearly indicates the level of classification involved in any description. No two kinds of animals have the same binomial name, and every animal has only one correct name, as required by the *International Code of Zoological Nomenclature*, thereby avoiding the confusion that common names cause. ^{ICZN}

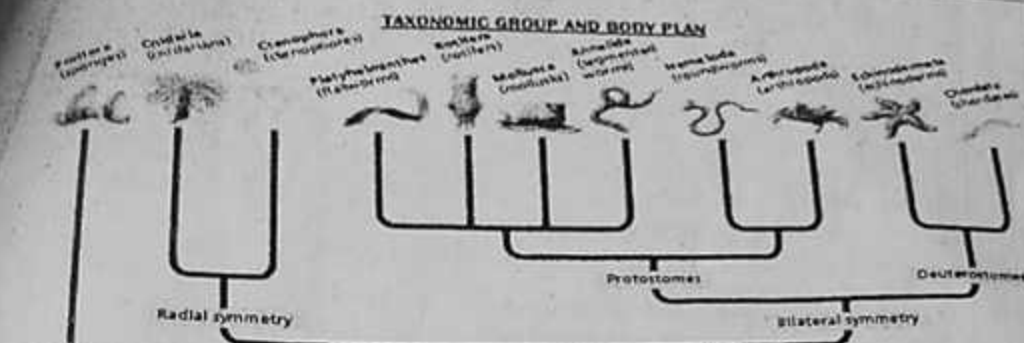
Generally, living organisms are classified into three domains and six Kingdoms as outlined below.

Three-domain system



Six-kingdom system

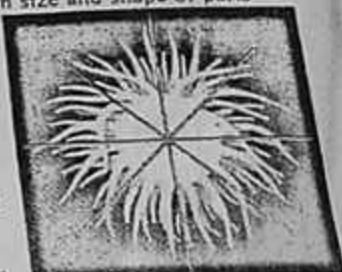




No true tissues
 Animals are unicellular or multicellular heterotrophic organisms that lack cell walls. Some animals, called vertebrates, have backbone. Other animals, called invertebrates, do not have a backbone. Unlike the cells of unicellular organisms, the cells of multicellular organisms do not lead independent lives. Each cell depends on the presence and functioning of other cells.

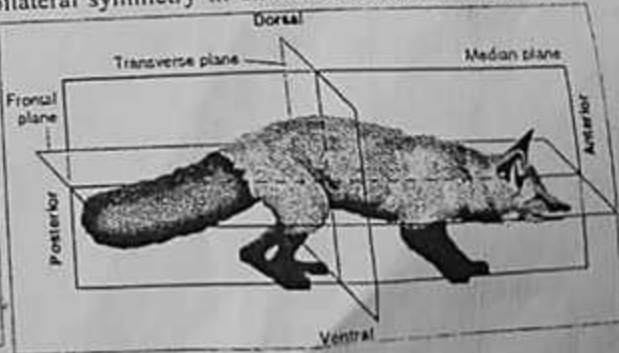
Body Plan (Symmetry, Embryonic cell layers and body cavities)

One of the most strikingly ordered series of changes in evolution is reflected in body plans in the animal kingdom and the protists. A body plan describes an animal's shape, symmetry and internal organization. Sponges (Porifera) have the simplest body plan of all animals. They are asymmetrical, which means that they do not display symmetry. Symmetry is a body arrangement in which parts that lie on opposite sides of an axis are identical. Symmetry may also refer to balanced proportions, or correspondence in size and shape of parts on opposite sides of a median plane. There are two divisions of symmetry:



- 1) **Radial symmetry:** This is a body plan in which the parts are organized in a circle around an axis. It applies to forms that can be divided into similar halves by more than two planes passing through the longitudinal axis. These are tubular, vase, or bowl shapes found in hydras, jellyfish, sea urchins, and related groups, in which one end of the longitudinal axis is usually the mouth. The two phyla that are primarily radial (Cnidaria and Ctenophora), are called the Radiata.

- 2) **Bilateral symmetry:** These are animals that can be divided into two mirrored portions- right and left halves. They have two similar halves on either side of a central plane. Most animals have a dorsal (back) and ventral (abdomen) side, an anterior (towards the head) and posterior (towards the tail) end, and a right and left side. The appearance of bilateral symmetry in animal evolution was a major advancement, because bilateral animals are much better fitted for directional (forward) movement than are radially symmetrical animals. Bilaterally symmetrical animals tend to exhibit cephalization (the concentration of sensory and brain structures in the anterior end of the animal). As a cephalized animal moves through its environment, the anterior end precedes the rest of the body, sensing the environment.



Cell layers and Body Cavities

Body parts are organized into layers derived from two embryonic tissue layers. Ectoderm gives rise to the epidermis, the outer layer of the body wall. Endoderm gives rise to the gastrodermis, the tissue that lines the gut cavity. Between the epidermis and the gastrodermis is a non-cellular layer called mesoglea. A third layer is sandwiched between the ectoderm and endoderm. This layer is mesoderm, which gives rise to supportive, contractile, and blood tissues.

Bilateral animals can be grouped according to the presence and type of body cavity. A body cavity is a fluid-filled space in which the internal organs can be suspended and separated from the body wall, called coelom. A major evolutionary innovation appearing within the Bilaterally symmetrical animals is the coelom. This is a fluid-filled space that surrounds the gut. The coelom provides "a tube within a tube" arrangement that allows much greater flexibility of the body cavity. The coelom also provides space for visceral organs. The fluid-filled coelom additionally serves as a hydrostatic skeleton in some forms, especially many worms, aiding in such activities as movement and burrowing.

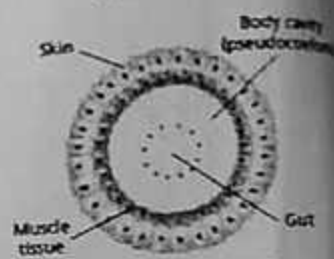
Animals such as flatworms do not have a body cavity and are called acoelomate. In these animals, the region between the ectodermal epidermis and the endodermal digestive tract is completely filled with mesoderm in the form of a spongy mass of space-filling cells called parenchyma. However, most animal phyla have body cavities that separate their digestive tract from the outer wall.

In some phyla, the mesoderm lines the interior of the coelom but does not surround the exterior of the endodermal gut. A cavity such as this, that is not completely lined by mesoderm is called pseudocoelom, which means false body cavity as seen in Nematodes (roundworms) and the animals that exhibit such are pseudocoelomate. A cavity completely lined by mesoderm is called a coelom.

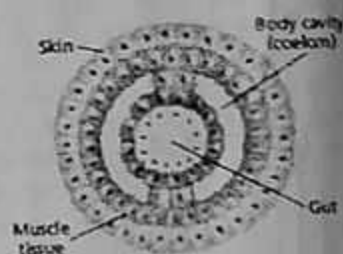
Animals that have coeloms are called coelomates. In coelomates, mesoderm lines the body cavity and supports the endodermal gut. The mesoderm also forms the tissues of attachment for the organs located in the coelom, such as liver and the lungs. Molluscs, annelids, arthropods, chordates and echinoderms are all coelomates.



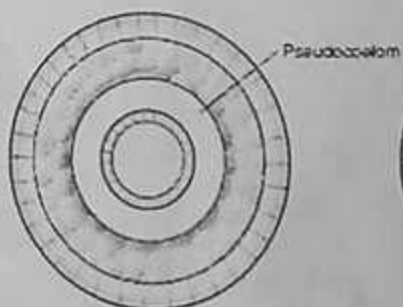
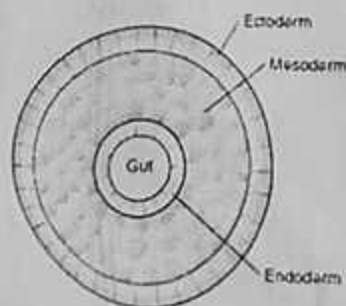
(a) ACOELOMATE



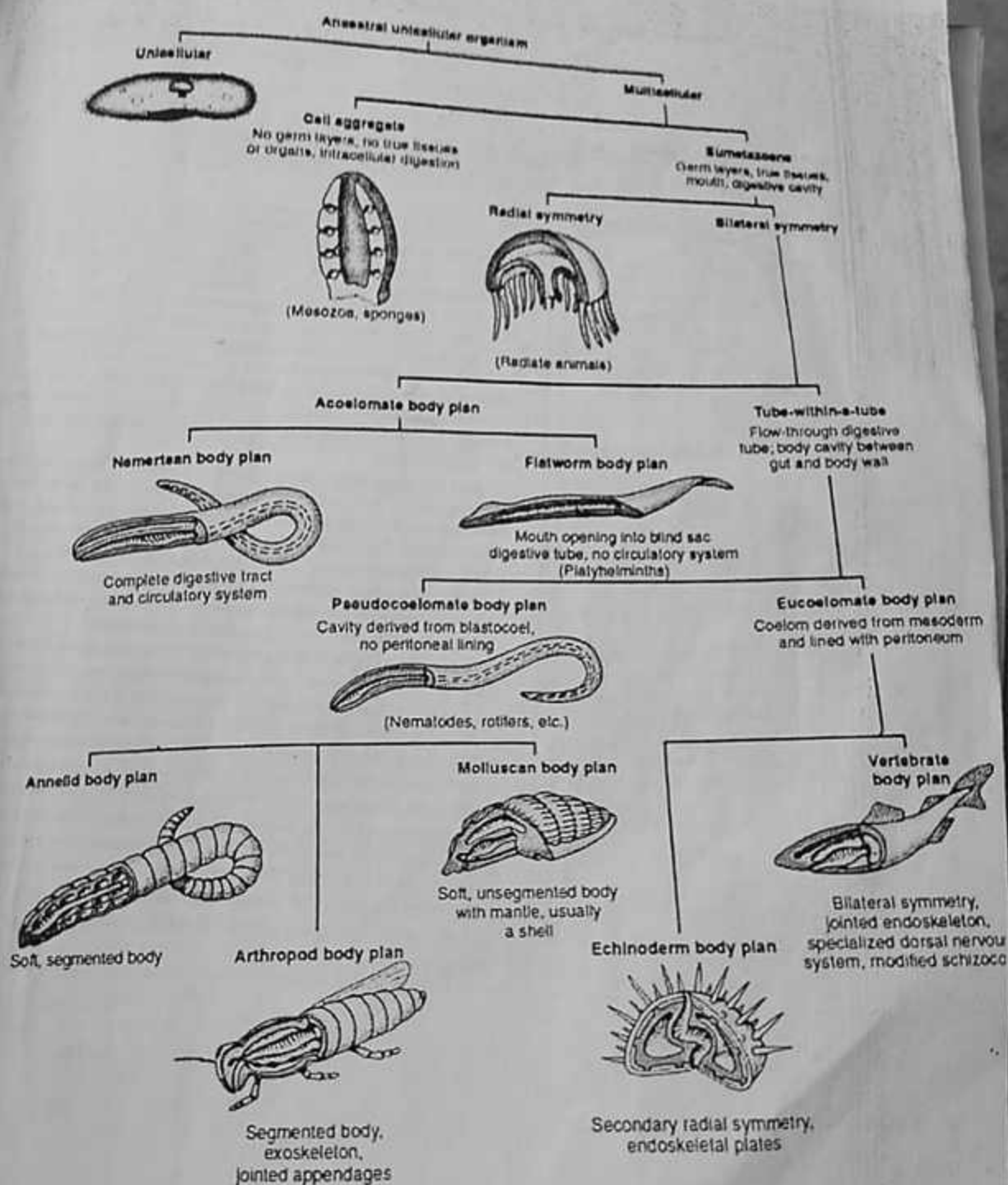
(b) PSEUDOCOELOMATE



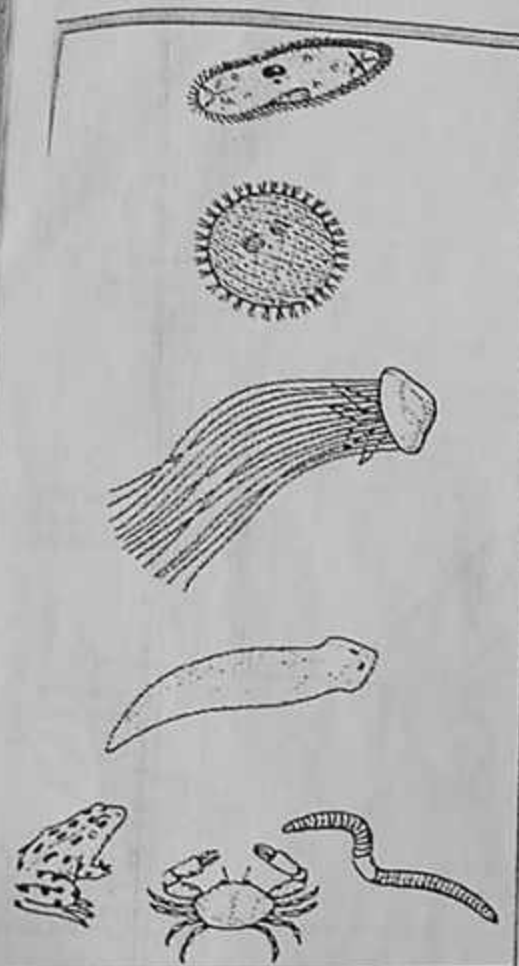
(c) COELOMATE



Legend:
 Ectoderm
 Mesoderm
 Endoderm



Levels of Organization in Animal Complexity



1. **Protoplasmic level of organization.** Protoplasmic organization is found in unicellular organisms. All life functions are confined within the boundaries of a single cell, the fundamental unit of life. Within a cell, living substance is differentiated into organelles capable of carrying on specialized functions.
2. **Cellular level of organization.** Cellular organization is an aggregation of cells that are functionally differentiated. A division of labor is evident, so that some cells are concerned with, for example, reproduction, others with nutrition. Such cells have little tendency to become organized into tissues (a tissue is a group of similar cells organized to perform a common function). Some protozoan colonial forms that have distinct somatic and reproductive cells might be placed at the cellular level of organization. Many authorities also place sponges at this level.
3. **Cell-tissue level of organization.** A step beyond the preceding is the aggregation of similar cells into definite patterns or layers, thus becoming a tissue. Some authorities assign sponges to this level, although jellyfishes and their relatives (Cnidaria) more clearly demonstrate the tissue plan. Both groups are still largely of the cellular grade of organization because most of the cells are scattered and not organized into tissues. An excellent example of a tissue in cnidarians is the nerve net, in which the nerve cells and their processes form a definite tissue structure, with the function of coordination.
4. **Tissue-organ level of organization.** Aggregation of tissues into organs is a further step in complexity. Organs are usually made up of more than one kind of tissue and have a more specialized function than tissues. This is the organizational level of the flatworms (Platyhelminthes), in which there are a number of well-defined organs such as eyespots, digestive tract, and reproductive organs. In fact, the reproductive organs are well organized into a reproductive system.
5. **Organ-system level of organization.** When organs work together to perform some function we have the most complex level of organization—the organ system. Systems are associated with basic body functions—circulation, respiration, digestion, and others. The simplest animals that show this type of organization are nemertean worms, which have a complete digestive system distinct from the circulatory system. Most animal phyla demonstrate this type of organization.

KINGDOM PROTISTA:

Sub-Kingdom PROTOZOA

The protozoa are large collection of organisms that have only one thing in common in that they are all unicellular (consist of a single cell, although some species within the group contain more than one nucleus during all or portions of their life cycles). The unicellular protozoan groups are the simplest animal-like organisms. They are nonetheless complete organisms that perform all basic functions of life seen in more complex animals. Within confines of their cell, they show remarkable organization and division of labour, possessing distinct supportive structures, locomotor devices, fibrils, and simple sensory structures. The diversity observed among unicellular organisms is achieved by varying the architectural patterns of sub-cellular structures, organelles, and the cell as a whole.

Protozoa are found wherever life exists. They are highly adaptable and easily distributed from place to place. They are all grouped together into a sub-kingdom which is divided into several phyla. They require moisture, whether they live in marine or freshwater habitats, soil, decaying organic matter, or plants and animals. They may be sessile or free swimming, and they form a large part of the floating plankton. Protozoa like all cells are covered by a plasma membrane. Many protozoa have more than one such membrane as part of their pellicle. Protozoa possess a great diversity of membranous organelles in their cytoplasm including Mitochondria and Golgi apparatus. They have well-defined chromosomes contained within a membrane-bound nucleus.

Main characteristics of the Protozoa

- Single celled or unicellular and possess membrane-bound nucleus.
- Entire body bounded by a plasmalemma.
- The cytoplasm is bounded by a clear outer gelatinous region the ectoplasm and an inner more fluid region the endoplasm.
- Within the cytoplasm are the organelles; nucleus (or in some species nuclei), nucleolus, chromosomes, Golgi bodies, endoplasmic reticulum, lysosomes, centrioles, mitochondria and in some cases chloroplasts.
- Organelles unique to protozoa include: contractile vacuoles, trichocysts and toxicysts. Contractile vacuoles are organelles involved in expelling water from cytoplasm, for volume regulation and osmotic regulation. Trichocysts take the form of elongated capsules triggered by mechanical and/or chemical stimuli. Toxicysts are related structures involved in predation- discharged filaments paralyse prey and initiate digestion.
- Reproduction may be either asexual or sexual, although many species alternate the two types in their life cycles or perform one or the other reproductive functions in response to environmental conditions. Most often asexual reproduction is by binary fission, in which one individual divides into two. Nuclear division during asexual reproduction is mostly by mitosis. Multiple fission (merogony, schizogony) occurs in some amoebas and in Apicomplexa. In this type of division the nucleus and other essential organelles divide repeatedly before cytokinesis, thus producing a large number of daughter cells almost simultaneously. Another type of multiple fission often recognized is sporogony, which is meiosis immediately after the union of gametes, typically followed by mitosis.
- Protozoa move chiefly by cilia, flagella or pseudopodia. *→ major type of locomotion*
- They exhibit every imaginable means of transmission from contamination to sexual and vector-assisted.
- Digestion is entirely intracellular- ingestion is by phagocytosis or endocytosis. Food particles become surrounded by external membrane forming a food vacuole which moves about within the cytoplasm. The food vacuole is first acidic and then becomes basic. Digestive enzymes are secreted into the vacuole. Soluble nutrients are absorbed into the endoplasm. Solid wastes are discharged through an opening in the plasma membrane.
- They exhibit diverse kinds of Animal Associations including mutualism (both partners benefit), commensalism (one partner benefits without affecting the other), or parasitism (one partner benefits at the expense of the other). Some of the most important diseases of humans and domestic animals are caused by parasitic protozoa.
- Nutrition is majorly heterotrophic (depending on other plants or animals for food) but some are saprozoic.

Classification of phylum Rhizopoda

Members of the sub-kingdom Protozoa may be classified into four different phyla with respect to commonly found organisms or parasites of human or veterinary importance.

1. **Phylum Rhizopoda:** These are the amoebas. They have pseudopodia for locomotion. The name Rhizopoda may be translated as "root-like foot" and refers to the process by which the cytoplasm flows within the cell to cause that are used for both movement and ingestion of food. Examples include *Amoeba proteus*. Some are parasitic amoebic dysentery, *Entamoeba histolytica* (the causative agent of constantly changing shape as they meet obstacles and change direction, or when they respond to environmental stimuli. Adverse stimuli such as acidic water or very bright light cause them to retreat, while favourable stimuli, such as contact with food, cause them to advance.



Typical structure of Amoeba



Amoeba surrounds a small flagellate with pseudopodia

1. Habitat of *Amoeba Proteus*

Amoeba Proteus is widely distributed. It is commonly found in freshwater pools or bottom mud, in ponds, ditches, lakes and slow streams, and often in shallow water on the underside of aquatic vegetation. It is also found in damp soils.

2. Structure of *Amoeba Proteus*

no cell wall

Amoeba proteus is a one-celled microscopic animal about 0.25 mm (250 microns) in diameter and so transparent that it is invisible to the naked eyes. Under the compound microscope, it appears as an irregular, colourless, translucent mass of living animal-like jelly or protoplasm that is constantly changing its shape by sending out and withdrawing finger-like processes, the pseudopodia.

The name *Amoeba* is derived from a Greek word *amoebe* which means change. The specific name *proteus* is based on the name of a mythological Greek sea-god who constantly changes his shape. Although it possesses no cell wall, it has a thin delicate outer membrane called the plasmalemma. Just beneath this is a non-granular layer, the ectoplasm which encloses the granular endoplasm. However, there is no line of demarcation between the ectoplasm and endoplasm.

Locomotion of *Amoeba Proteus*

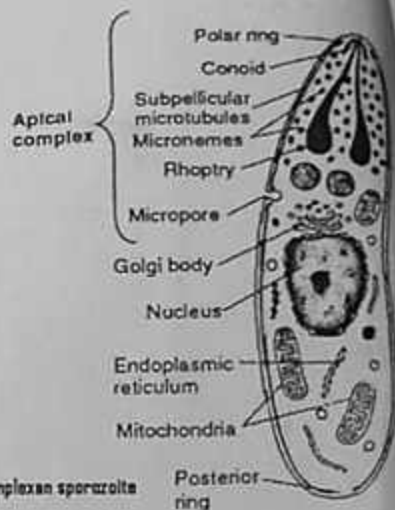
Amoeba Proteus exhibits characteristic amoeboid movement by the formation of finger-like temporary processes, the pseudopodia. The pseudopodia of *Amoeba* are known as *lobopodia* due to their blunt, finger-like appearance and rounded tips. The ectoplasm forms a blunt projection into which the endoplasm flows to form a pseudopodium at the advancing end which is then spoken as the anterior end. Usually several small pseudopodia are formed at first, one of these enlarges and the others disappear. The protoplasm which enters pseudopodia is naturally withdrawn from other parts, so that the animal not only changes its body shape but also its position, thus, pseudopodia bring about change of shape and position of the

animal. These are called amoeboid movements which occur not only in Amoeba but in other Protozoa and some amoeboid cells of Metazoa.

4. Nutrition in *Amoeba Proteus*

An Amoeba is unable to form its food from simple substances, but it requires ready-made organic substances for food; such a mode of nutrition in which solid organic particles are ingested is called zootrophic or holozoic. The food of Amoeba consists of algal cells and filaments, bacteria, other protozoans, small metazoans such as rotifers and nematodes and organic matter. Smaller flagellates and ciliates appear to constitute the favourite food of Amoeba. There is no mouth but food is ingested at any point which is generally at the anterior advancing end. The nutrition involves a number of processes, viz., ingestion, digestion, assimilation, dissimilation and egestion.

Phylum Apicomplexa: Apicomplexans are organisms that possess a certain combination of structures, called an **apical complex**, distinguishable only by electron microscopy. All apicomplexans are endoparasites, and their hosts are found in many animal phyla. They all have a single type of nucleus and no cilia or flagella, except for the flagellated microgametes in some groups. The presence of a certain combination of organelles, the apical complex, distinguishes this phylum (see Figure). The apical complex is usually present only in certain developmental stages of the organisms; for example, merozoites and sporozoites. Some structures, especially the rhoptries and micronemes, apparently aid in penetrating the host's cell or tissue. Examples include *Plasmodium falciparum*, *Toxoplasma gondii*, *Plasmodium ovale*, *Cryptosporidium parvum*, *Plasmodium malariae*, *Eimeria tenella*, *Plasmodium vivax*.



Typical structure of Apicomplexan sporozoite

Life Cycle of *Plasmodium falciparum*

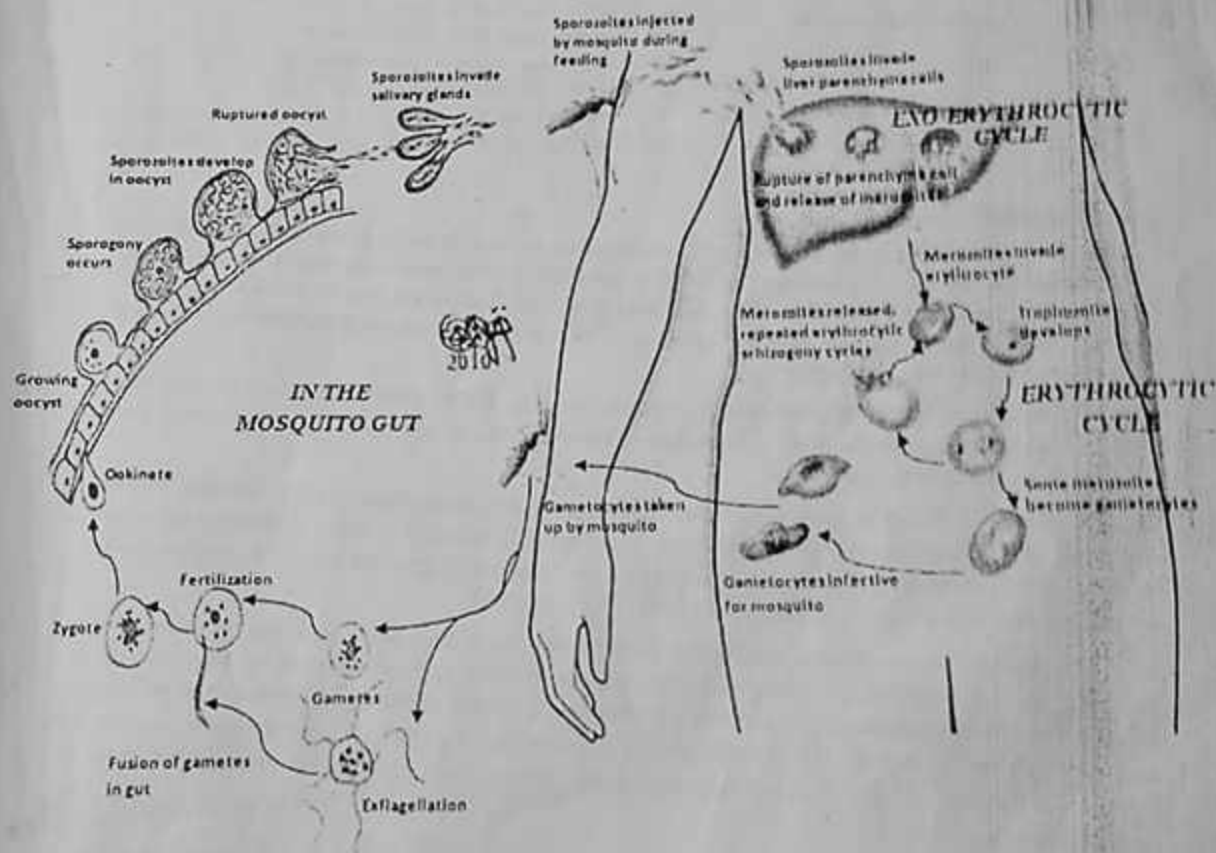
The *Plasmodium* species that infect humans have a complex life cycle that involves both asexual and sexual reproductive stages. They are all transmitted by anopheline (female *Anopheles*) mosquitoes and multiplication takes place in both humans and the mosquito vector. Only female mosquitoes feed on blood, as they require the nutrients contained in it to produce eggs and therefore, they are the ones that transmit the disease.

Humans become infected when they are bitten by infective female *Anopheles* mosquito (harbouring sporozoites) during blood meal. Sporozoites are deposited through the skin to the blood vessels. Once they have gained access to the bloodstream, the sporozoites are transported to the liver where they penetrate the hepatocytes (liver cells). Within the hepatocytes, the sporozoites undergo morphological change (losing its apical complex and surface coat) and multiply asexually in a process called exoerythrocytic schizogony to form thousands of merozoites. Schizogony is a form of asexual reproduction in which the nucleus divides several times and then the parent cell divides into as many individual merozoites as there are nuclei. The infected hepatocyte then ruptures, releasing the merozoites into the bloodstream.

The merozoites then invade the red blood cells in which they transform into another morphological form called the trophozoites and these also reproduce asexually by schizogony. Because these divisions take place in the red blood cells, it is referred to as erythrocytic schizogony. Once inside an erythrocyte, the merozoite develops into a feeding trophozoite which ingests host cytoplasm and forming a large food vacuole (parasitophorous vacuole) round the trophozoite and the trophozoite takes on a 'signet-ring' shape, more or less amoeboid and uninucleate.

The young trophozoite grows substantially before undergoing several nuclear divisions. This is soon followed by the division of cytoplasm forming a schizont, each containing merozoites in numbers characteristic of the parasite species. The merozoites then invade fresh erythrocytes and the cycle continues. The erythrocyte eventually bursts releasing the merozoites into the circulation and these then invade non-infected red blood cells. The result is a dramatic increase in parasitaemia and, when the number of parasites reaches a certain level, the process of schizogony and the subsequent liberation of merozoites become synchronized and periodic. The length of the intra-erythrocytic development is variable, being 48hr for *P. falciparum*, *P. vivax* and *P. ovale*, and 72hr for *P. malariae*. At the end of each cycle when the erythrocytes burst and release the merozoites into the bloodstream, 'toxins' are released into the host. Each time there is the liberation of parasites into the bloodstream it has a pyrogenic effect upon the host, making the host to feel feverish. After a few erythrocytic cycles, some merozoites develop into a male or a female gametocyte. Whether this development is pre-determined genetically or as a response to some specific stimulus is unknown.

During a blood meal from an infected human, the female *Anopheles* mosquito takes up gametocytes, which transform into gametes. The maturation of the female gametocyte into a macrogamete takes place without major morphological changes. In the extracellular male gamete, the nucleus divides three times and each of the eight nuclei formed combines with cytoplasm to form 6-8 thread-like microgametes in exflagellation, resulting in the formation of a diploid zygote. This soon elongates to become a motile ookinete. The ookinete crosses the peritrophic membrane of the midgut, then passes through the single cell layer of the midgut epithelium and establishes itself beneath the basal lamina, forming an oocyst. The oocyst gradually increases in size, enlarges and undergoes nuclear division forming sporoblasts. The sporoblasts in turn divide repeatedly to form thousands of sporozoites. Being motile, the sporozoites burst through the weakened or ruptured wall of the oocyst and invade the body cavity of the mosquito. From the haemolymph, sporozoites migrate to the salivary glands of the female *Anopheles* mosquito, which becomes infective. When the mosquito feeds on blood meal, it then injects sporozoites into the bloodstream of the vertebrate host.

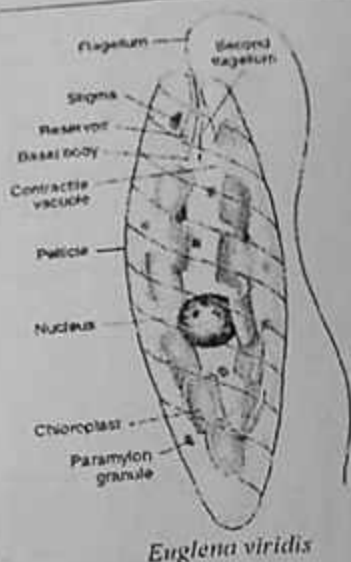


Life Cycle of the Human Malaria Parasite

- iii. **Phylum Kinetoplastida (Mastigophora):** The Phylum Kinetoplastida contains species that parasitize a wide range of organisms from humans to plants. Members of this group are characterized by the possession of a basal body (the *kinetosome*), at the base of the flagellum. Examples include *Euglena viridis*, *Trypanosoma brucei*, *Trypanosoma cruzi*, *Trypanosoma gambiense*, *Trypanosoma congolense*, *Leishmania donovani*, *Leishmania tropica*.



Typical structure of Trypanosoma



Euglena viridis

Euglena viridis

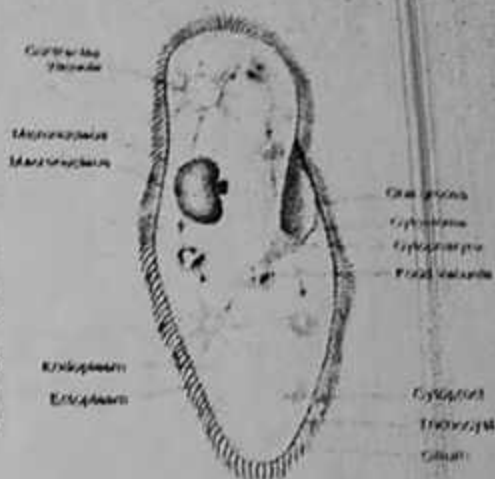
Euglena viridis (in the figure above- right) is a flagellate commonly studied protozoan organism. Its natural habitat is freshwater streams and ponds where there is considerable vegetation. The organisms are spindle shaped and about 60 μm long, but some species of *Euglena* are smaller and some larger (*E. oxyuris* is 500 μm long). Just beneath the outer membrane of *Euglena* are proteinaceous strips and microtubules that form a pellicle.

In *Euglena* the pellicle is flexible enough to permit bending, but in other euglenids it may be more rigid. A flagellum extends from a flask-shaped reservoir at the anterior end, and another, short flagellum ends within the reservoir.

A kinetosome is found at the base of each flagellum, and a contractile vacuole empties into the reservoir. A red eyespot, or stigma, apparently functions in orientation to light. Within the cytoplasm are oval chloroplasts that bear chlorophyll and give the organism its greenish color. Paramylon bodies of various shapes are masses of a starchlike food storage material.

Nutrition of *Euglena* is normally autotrophic (holophytic), but if kept in the dark the organism makes use of saprophytic nutrition, absorbing nutrients through its body surface. Mutants of *Euglena* can be produced that have permanently lost their photosynthetic ability. Although *Euglena* does not ingest solid food, some euglenids are phagotrophic. *Peranema* has a cytostome that opens alongside its flagellar reservoir. *Euglena* reproduces by binary fission and can encyst to survive adverse environmental conditions.

- iv. **Phylum Ciliophora:** Ciliates are a large and interesting group, with a great variety of forms living in all types of freshwater and marine habitats. They possess simple cilia or compound ciliary organelles in at least one stage of their life cycle. Cilia are generally similar to flagella, except that they are much shorter, more numerous, and widely distributed over the surface of the protozoan. Most ciliates are free living, but many are commensals of vertebrates and invertebrates, and a few are parasitic. They are usually solitary and motile, but some are sessile and some colonial. Ciliates are always multinucleate, possessing at least one **macronucleus** and one **micronucleus**. Examples include *Balantidium coli*, *Paramecium caudatum*, *Paramecium aurelia*, *Vorticella* spp.



Typical structure of a Paramecium

Home Work

Explain how a **named protozoa** eat, digest food, osmoregulate, and respire.

Submit on Monday, 24th September, 2018

Phylum Annelida

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❖ INTRODUCTION

- ANNELIDA = Annelus "little ring" + ida "worms"
- The phylum Annelida consists of worms whose body are divided into similar segments (also called metamerer)
- The most distinguishing characteristic that sets them apart from other wormlike creatures is their segmentation.
- This repetition of body parts is also called metamerism
- This phylum consists of approximately 15,000 species.
- The members of this phylum have a worldwide distribution, though few species are cosmopolitan.
- The most familiar of these members are the earthworms and fresh water worms.

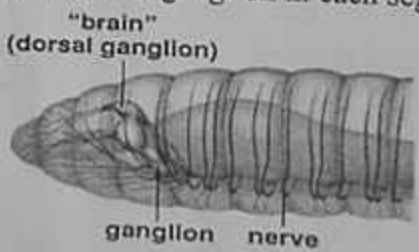
❖ GENERAL CHARACTERISTICS OF ANNELIDS

- The body is vermiform, bilaterally symmetrical, and metamerically segmented
- Each segment is separated from contiguous ones by a transverse septum
- The body is triploblastic with a well developed coelom
- The body is covered with a flexible non-chitinous cuticle
- Body wall with well developed outer circular and inner longitudinal layers of muscle with glandular epidermis.
- Possession of chitinous chatae which are borne on parapodia in some.
- Possession of COMPLETE DIGESTIVE SYSTEM.
- Respiration by skin or gills
- Excretory system consists of segmentally arranged nephridia.
- Well developed nervous system
- May be hermaphrodite or of separate sexes
- Presence of a trocophore larva in some, others have direct development.
- Closed blood vascular system with haemoglobin.

❖ BIOLOGY OF ANNELIDS

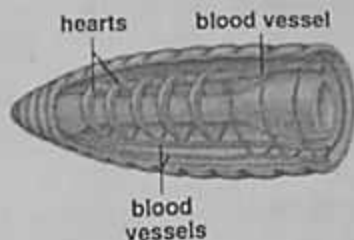
Nervous System

- Consists of a brain, which is connected to a pair of ventral longitudinal nerve cords, with a ganglion in each segment (metameric)



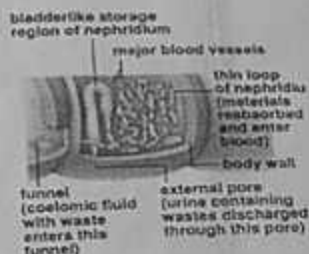
❖ Circulatory System

- **Closed circulatory system**, in which the blood is always enclosed within blood vessels that run the length of the body and branch to every segment
- Several hearts (5 in earthworms) are used to pump blood through the closed circuit



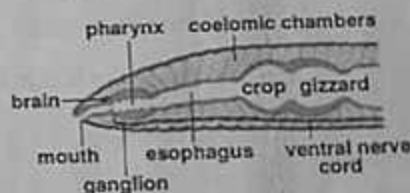
❖ Excretory System

- Consists of paired (metameric) **metanephridia**
- Excretory tubes with ciliated funnels that remove waste from the coelomic fluid; open to the outside via excretory pores.



Note:

- Not all organ systems are metameric
- For example, the digestive system extends the length of the organism and is differentiated along its length



❖ Reproductive System

- Most annelids are hermaphroditic, but they are usually cross fertilizers.
- Earthworms and leeches form pairs and reciprocally fertilize one another
- Some annelids (e.g. marine sandworms) are dioecious and they release eggs and sperm into the marine environment, where gametes unite to form trochophore larvae

❖ Classes of Annelida

- Class Polychaeta (fanworms, clam worms)
- Class Oligochaeta (earthworms)
- Class Hirudinea (Leeches)

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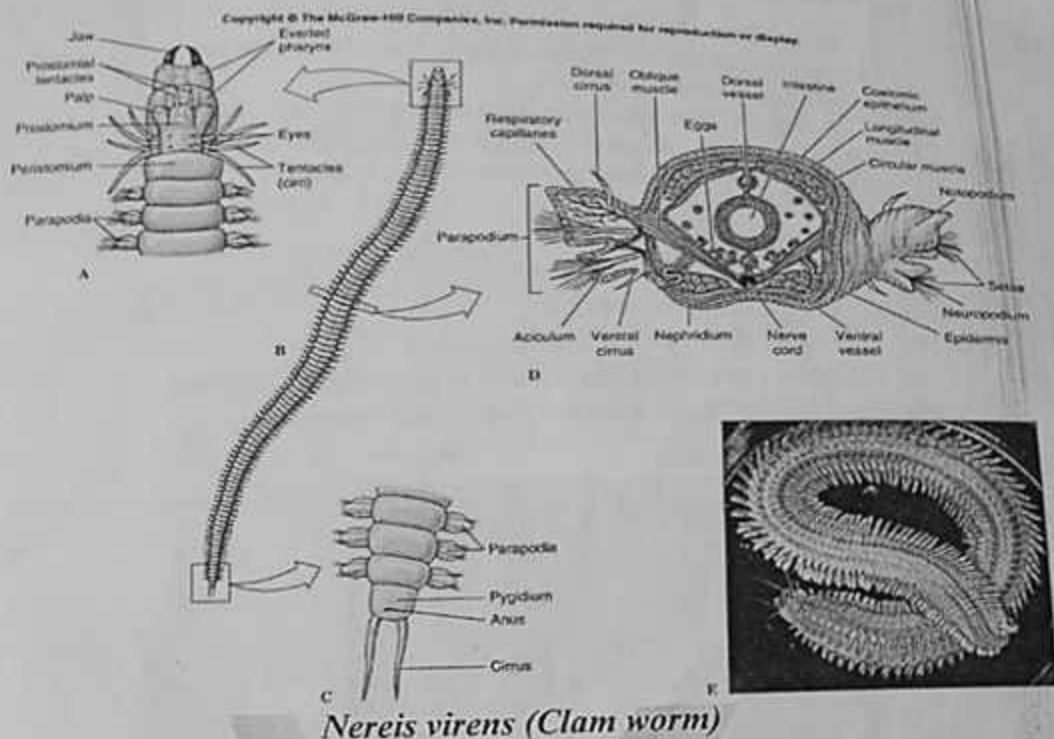
❖ Class - Polychaeta ("many bristles")

- Marine worms, including, Arenicola (Lug worm), Amphirite (plumworm), chaetopterus and Nereis (clamworms).
- Each segment is equipped with a pair of - **parapodia**; used in locomotion
- Parapodia contain a large number of chitinous bristles called - **setae**; used in anchoring the worms
- Body typically has two-part head composed of a **prostomium** and a **peristomium**, followed by a segmented body and a terminal portion called the **pygidium** carrying the anus.
- **Prostomium** is well equipped with sensory and feeding structures



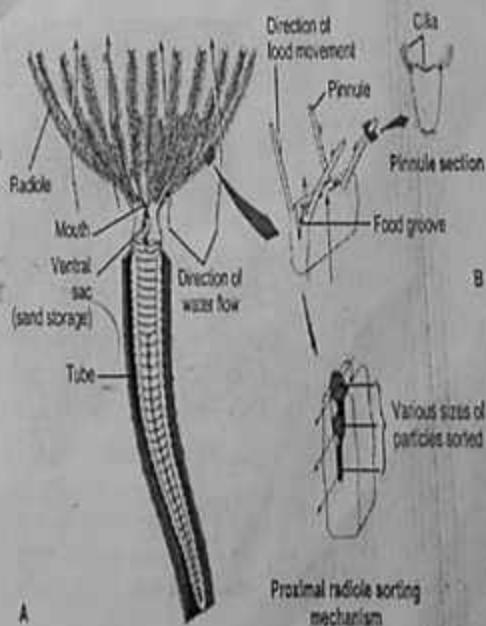
- Mouth is located just below the prostomium, but in front of the modified segments – **peristomium**
- Digestive system includes a muscular pharynx that can be everted through the mouth
- Pharynx is equipped with pincer-like jaws
- Although many of the smaller polychaetes lack respiratory structures, the larger one do possess gills
- Gills are usually modifications of the parapodia





❖ Class Polychaeta: Diversity

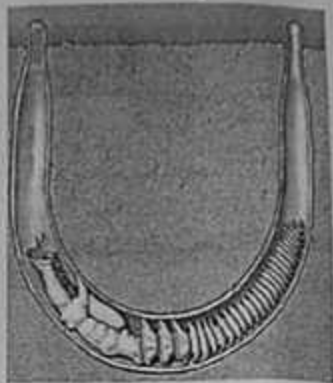
- Although a number of polychaetes are active predators, some are sedentary and burrow into mud or live in protective tubes in the mud
- In several of these species filter feeding has evolved
- A good example is the fan worm *Sabella*, with their feather-like head structures called **radioles**



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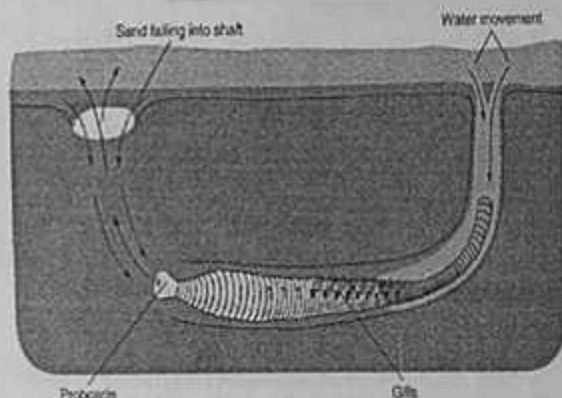
❖ Class Polychaeta: Diversity cont.

- *Chaetopterus* is tube dweller; lives in a U-shaped tube
- Parapodia are highly modified into 3 fan-like structures that bring water into the tube
- The **notopodium** secretes a mucous bag that traps food from the water flowing through the tube; the bag is periodically passed anteriorly toward the mouth



❖ Class Polychaeta: Diversity cont.

- *Arenicola* lives in a J-shaped burrow
- It employs peristaltic movements to generate a water flow
- Food is filtered out from the front of the burrow

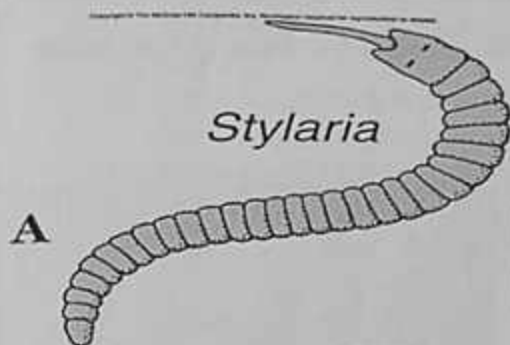


❖ Class Oligochaeta ("few bristles")

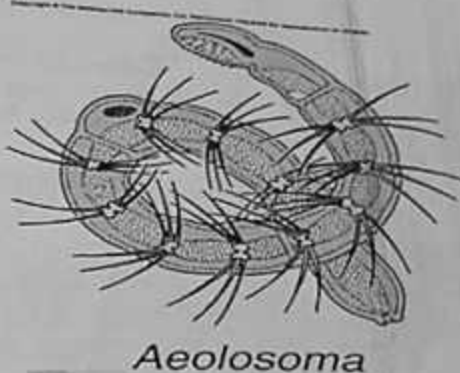
- Members are terrestrial with a few freshwater species
- Many of the morphological structures are reduced when compared to the polychaetes
- No distinct head appendages
- Clitellum present
- Prostomium lacks sensory structures
- Parapodia are absent; each segment usually contains one or more pairs of setae; used in locomotion
- Aquatic forms usually have larger setae than the terrestrial forms
- Examples are stylaria, dero, aeolosoma and earthworm

❖ EXAMPLES OF OLIGOCHAETAS

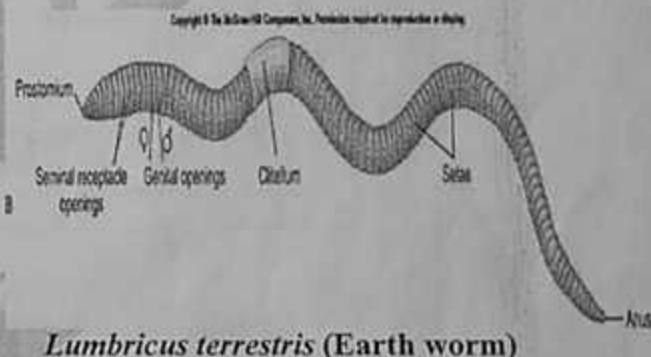
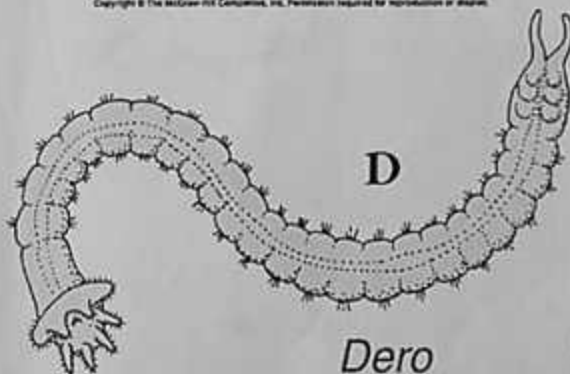
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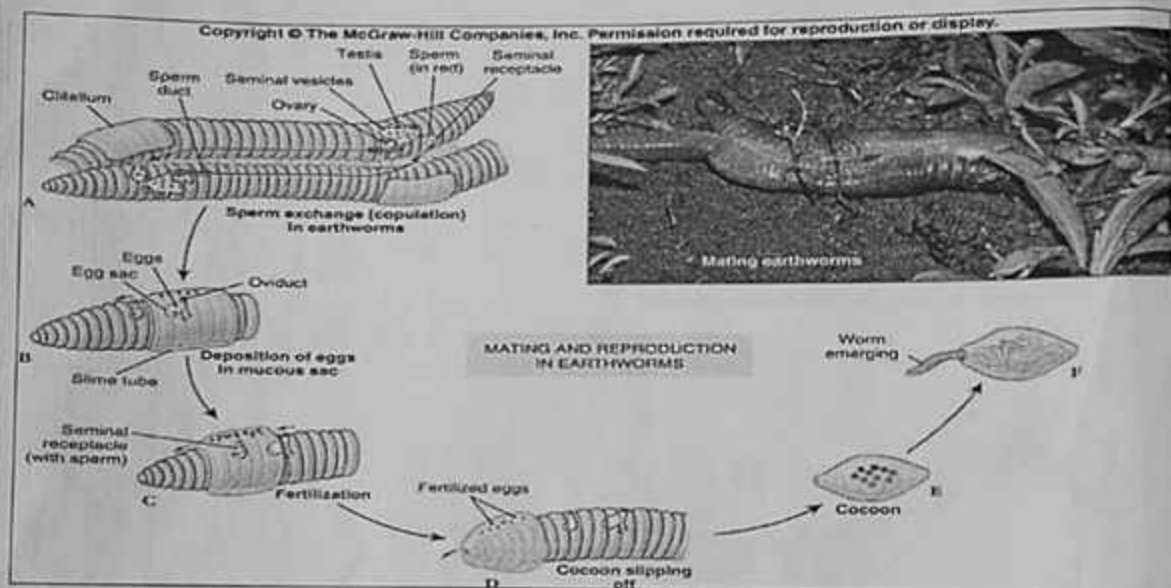
❖ BIOLOGY OF A TYPICAL OLIGOCHAETE (EARTHWORM)

Earthworms feed on vast quantities of soil that contains living and decaying organic material.

- Digestive tract of the annelids shows specialization along its length: mouth, pharynx, **crop** (food storage), **gizzard** (grinding), **calciferous glands** (accessory glands that excrete excess calcium from the food)
- Remainder of the gut is the intestine - for digestion and absorption
- Its surface area is increased because of a dorsal longitudinal fold called the **typhlosole**
- Lack respiratory organs; gas exchanges occurs across the body wall
- Hermaphroditic, but exchange sperm during copulation
- During copulation, worms join their anterior ends; held together by mucous secretions from a **clitellum**

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- After reciprocal copulation, sperm is stored in seminal receptacles
- Clitellum then secretes a mucous tube that serves as a cocoon
- The cocoon moves anteriorly and eggs from the oviduct and sperm from the seminal receptacles are poured into it; fertilization occurs in the cocoon
- Cocoon eventually slips off the anterior end of the worm. In time, young worms emerge from the cocoon



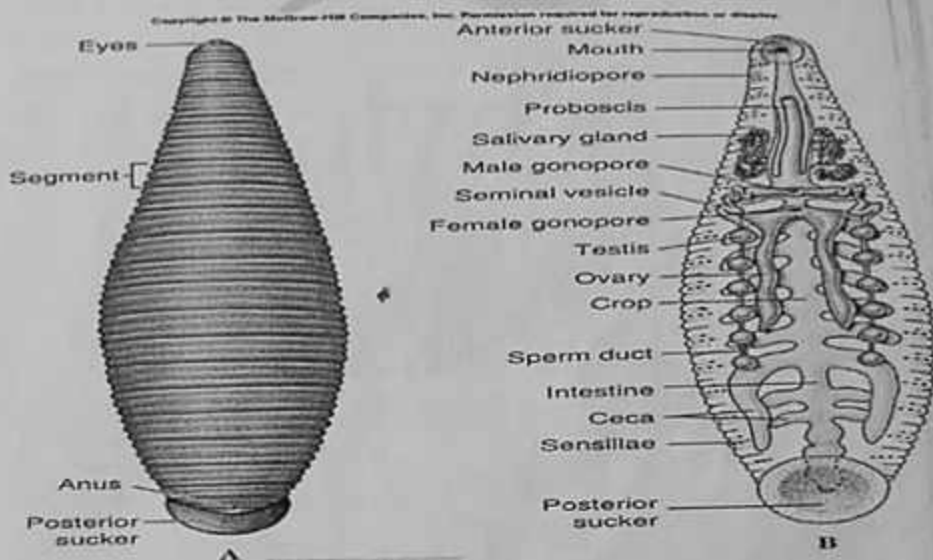
Mating and Reproduction in Earthworm

❖ Class Hirudinea

- Body is dorso-ventrally flattened
- Anterior segments are modified as a small sucker which surrounds the mouth; posterior segments form a larger sucker
- They are ectoparasites
- Setae are completely absent
- Evidence of segmentation externally which is subdivided into annuli, but no internal septa
- There is serial repetition of many of the organs (e.g., nephridia and testes)
- No parapodia or chaeta
- They are hermaphrodite, direct development.
- They develop clitellum when sexually mature.
- They have direct development.

❖ BIOLOGY OF A TYPICAL HIRUDINIDAE (LEECHES)

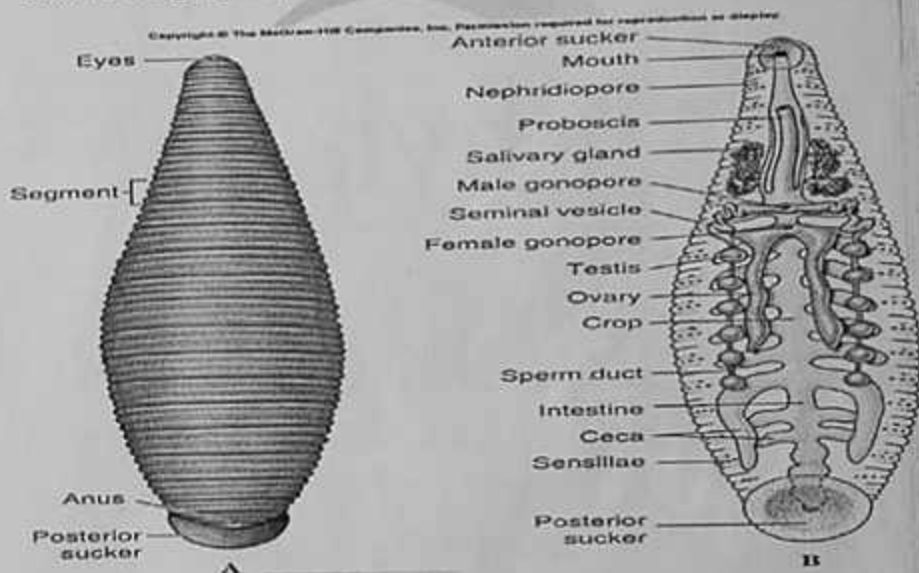
- Leeches are predominantly in freshwater habitats, but a few are marine
- They crawl over the surface in a loop like fashion, with the use of 2 suckers.
- Body is extended due to circular muscle contraction and the attachment of the anterior sucker to the substrate.
- Posterior sucker is subsequently released and longitudinal muscles contract bringing the posterior part of the body forward.



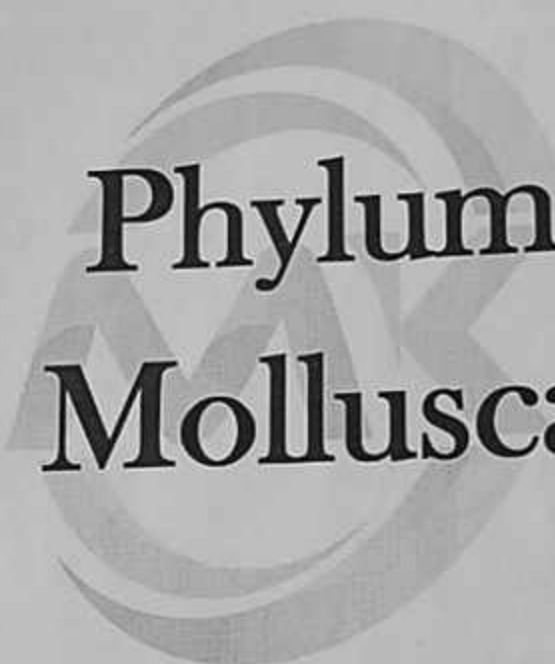
- Most leeches are active predators; however, some are the parasitic, bloodsucking forms.
- Blood suckers have blade like jaws that they use to penetrate the skin of a host.
- Blood is prevented from clotting because they secrete a powerful anticoagulant; anesthetics are also released
- A muscular pharynx subsequently pumps blood into the gut.
- Leeches are hermaphroditic but engage in cross-fertilization; some use hypodermic impregnation
- Leeches have a clitellum and are capable of generating a cocoon

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Phylum Mollusca

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❖ Phylum Mollusca

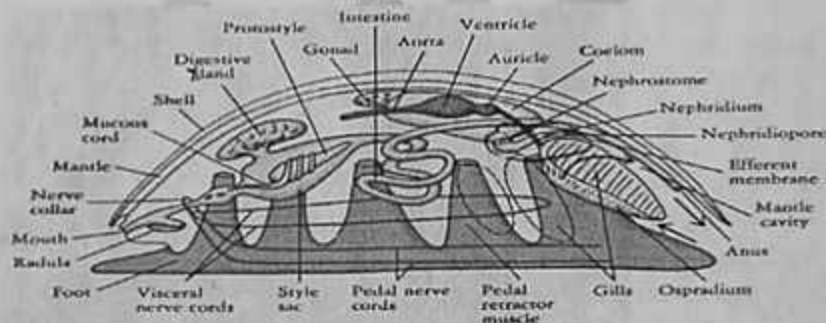
- Mollis = soft ; Mollusca = soft bodied animal
- Phylum Mollusca includes animals such as squids, snails, oysters, clams and slugs.
- Most are marine, but many are freshwater and some live on the land
- All members of this group have the same basic body plan, this is often indicated by presenting a **hypothetical ancestral mollusk (HAM)**

❖ HAM (hypothetical ancestral mollusc)

The **foot** - a broad, flat muscular organ that is adapted for locomotion and attachment

The **visceral mass** - contains the internal organs

The **mantle** - a fold of tissue that drapes over the visceral mass; space between the mantle and the visceral mass is called the **mantle cavity**



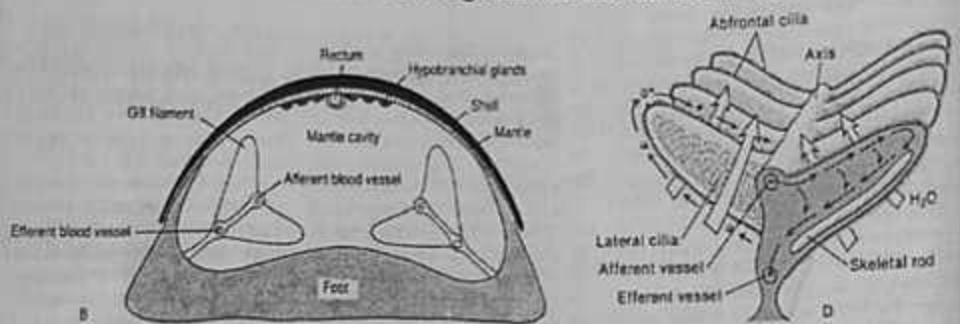
Hypothetical Ancestral Molluscs

❖ The Shell

- The mantle is responsible for secreting the shell.
- The shell is comprised of three layers:
 - The outside of the shell is covered by an organic layer - **periostracum**
 - The middle **prismatic layer** is characterized by densely packed prisms of calcium carbonate laid down in a protein matrix
 - The **inner nacreous layer** is composed of calcium carbonate sheets laid down over a thin layer of protein

❖ Gills

The gills of HAM are often indicated as one or more pairs of bipectinate gills, - flattened filaments attached to a longitudinal axis on either side



❖ The Radula

- The mouth cavity of HAM possesses a specialized rasping organ called the **radula**; sits on a cartilaginous structure – **odontophore**
- Particles of food brought into the mouth are bound in mucus secreted by the salivary glands

❖ Other Features of HAM

- Nervous system consists of a **nerve ring** and 2 **longitudinal nerve cords**
- Coelom is reduced
- **Open circulatory system**
- The excretory organs of the mollusks are **metanephridia**; inner ends open into the coelom via a ciliated funnel called the **nephrostome**; wastes leave the body via the **nephridiopore**

❖ General Characteristics of Molluscs

- 50,000 or more species of molluscs.
- They share 3 major sets of characteristics:
 - Body enclosed by a blanket-like mantle that secretes a shell made of calcium or some other stiff structure.
 - A mantle cavity between the mantle and the internal organs; the anus, reproductive and excretory ducts open into the mantle cavity.
 - A ventral, muscular foot that is highly modified among the various groups of molluscs.
- Free-living, multicellular animals
- Invertebrate (lack a backbone)
- Un-segmented

- Have a true coelom
- Bilateral symmetry
- Have a true heart
- Simple to complex nervous system

❖ Habitats & Niches

- Some Molluscs habitats include:
 - Marine (mostly)
 - Freshwater (a few)
 - Land (a few)
- Mollusc niches (life-style):
 - Most live on the ocean bottom and are mostly sedentary
 - Some are free -swimming (the cephalopods)

❖ CLASSIFICATION

- Aplacophora (a=without)
 - Ex. Neomenia
- Monoplacophora (mono=single; placo =shell; phora=bearing)
 - Ex. Neopilina
- Polyplacophora (poly=many)
 - Ex. Chiton
- Gastropoda (gastro=body cavity; poda=foot)
 - Ex. Common garden snail, abalone, turban snail
- Bivalvia(bi=two; valvia =valves)
 - Ex. Clam, mussels, union, Oyster
- Cephalopoda(cephalo=dealing with head; poda=foot)
 - Ex. Squid, octopus, etc.

❖ Class Aplacophora

➤ Specific characteristics:

- Wormlike body
- No shell, but has a body wall made up of calcium spicules (think of the sponges).
- It is this that puts them in the phylum mollusca
- They are generally found on the ocean floor.



❖ Class Monoplacophora Neopilina

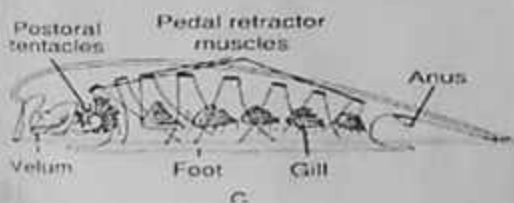
➤ Neopilina is the only genus in this entire class.

➤ Their specific characteristics include:

- Single shell
- Broad, rounded foot
- Evidence of segmentation

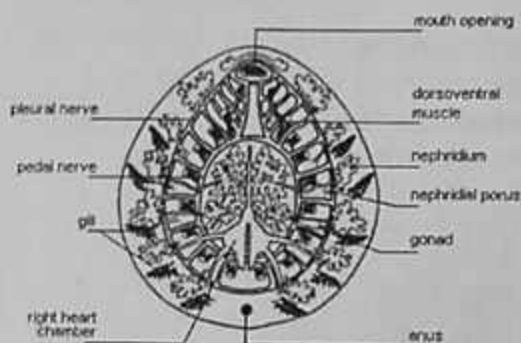
➤ They have 5-6 pairs of gills & eight pairs of foot retractors.

➤ This segmentation suggests an ancestral relationship with annelids.



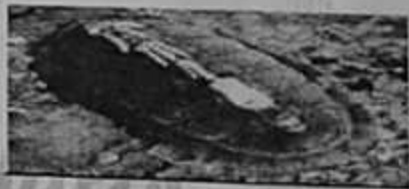
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❖ Class Monoplacophora



❖ Class Polyplacophora

- Chiton are the only genus in the class Polyplacophora
- Their specific characteristics:
 - Posses a shell that consists of 8 overlapping plates.
 - Thick mantle
 - Broad foot
 - Distinguishable head
 - They have a tongue like tooth covered rasping organ (radula) to scrape algae and other food.



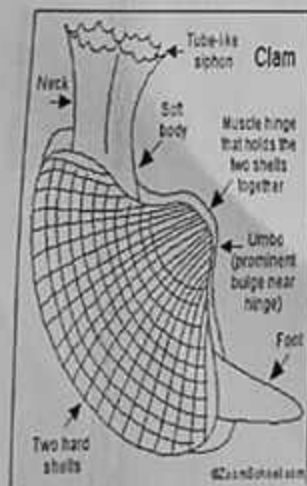
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❖ Class Gastropoda

- Most popular class of mollusks
- Consists of snails, helix and snail-like creatures.
- Specific characteristics:
 - They have a dorsally located shell (often coiled).
 - They have a well developed radula
- Visceral mass (organs) is located inside the shell. Their visceral mass is rotated 180 degrees during development (i.e. torsion).
- They are herbivores or predatory (carnivorous).
- Habitats include: ocean, lake, river bottoms, coastal shores, and land.



❖ Class Bivalvia



➤ Specific Characteristics:

- They have no head.
- They have two shells held together by powerful muscles.
- They have a ventrally located foot that sticks out between the two valves.
- Habitats include: marine and freshwater.
- They tend to burrow into soft mud or sand or attach to rocks or other shells.

❖ Class Cephalopoda

- Contains the largest molluscs Typical examples include logilo, sepia, octopus

➤ Specific Characteristics:

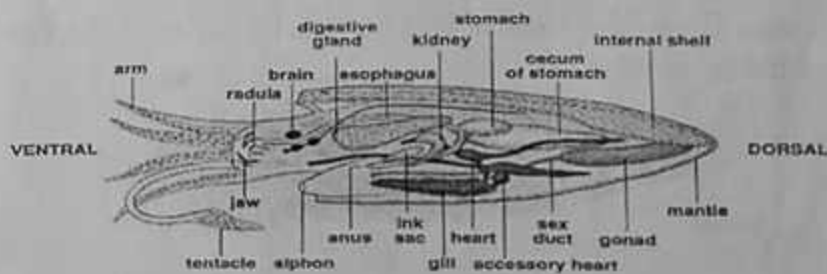
- May lack a shell (like an octopus)
- Shell may be reduced to a stiffening rod (like the squid or nautilus)
- The foot is highly modified to form a group of tentacles around the mouth.
- They are found in deep and shallow waters along many coasts.
- Squids & Nautilus are free-swimming and move very quickly. Octopuses are found among rocks or crawling on the bottom of the ocean.
- Complex brain, two lateral eyes, excellent eyesight

❖ Feeding among the cephalopods

- Cephalopods are carnivores
- Have a powerful parrot like beak that is used to tear prey apart.
- They also have a powerful radula
- In some of the octopuses the salivary glands are modified poison glands

❖ Locomotion

- Cephalopods are excellent swimmers: streamlined body; tentacles and fins as stabilizers
- Swim by means of jet propulsion, using the highly modified muscular mantle and the siphon
- By relaxing the mantle the mantle cavity is expanded and water can be drawn in
- By contracting the mantle water can be forced out of the mantle cavity by means of the small siphonal opening



❖ Shell

- Primitively the cephalopods possessed a shell; the fossil record indicates both coiled and non-coiled shells
- Extant members with coiled shells include Nautilus
- Some cephalopods (cuttlefishes) have an internal shell -cuddle bone
- The octopods have lost the shell entirely



❖ Other General Features

- For protection, they possess an ink sacs
- Cephalopods have well-developed sense organs, including a camera type eye
- Some have well-developed brains and show a remarkable capacity for learning.

- Cephalopods are the only molluscan class with a closed circulatory system

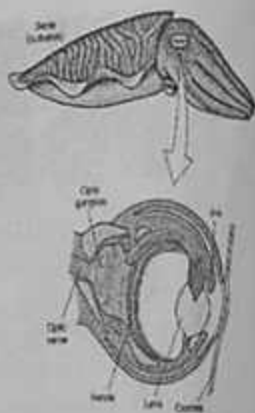
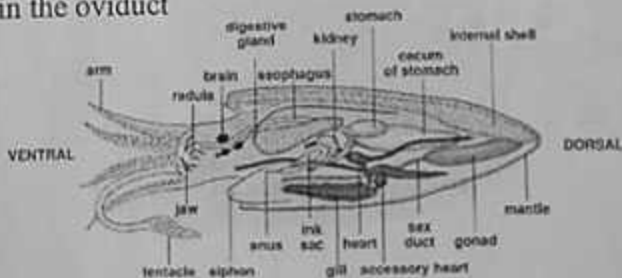


FIGURE 9.33

Eye of a cuttlefish (Sepia). The structure of a cephalopod eye shows a high degree of convergent evolution with the eye of vertebrates.

❖ Reproduction

- Sexes are separate
- Sperm is transferred to females in packets -spermatophores
- Male uses a tentacle to reach into its mantle cavity and pick up some spermatophores
- It then inserts the tentacle into the mantle cavity of the female near or within the oviduct



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❖ Typical Example of a Cephalopod

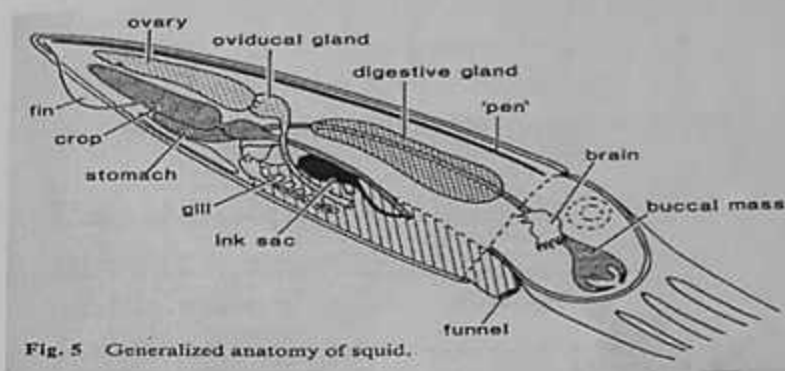
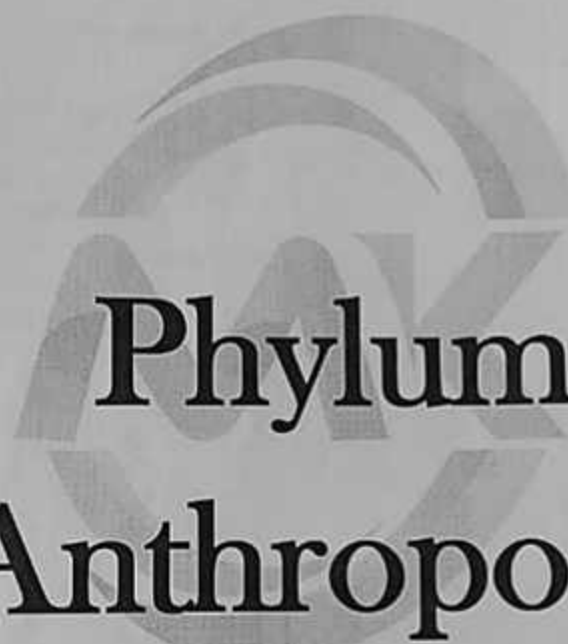


Fig. 5 Generalized anatomy of squid.

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Phylum Anthropoda

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❖ Overview

- Arthropod means "jointed appendages"
- Arthros and podus = Jointed and Legs
- Phylum arthropod is the largest phylum in the animal kingdom
- Most successful animal group
- Consists of over 1 million *named* species
- Members of these phylum inhabits all ecosystems

❖ GENERAL CHARACTERISTICS

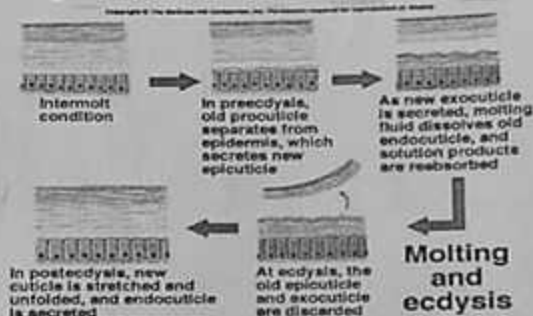
1. Triploblastic, coelomate, metamerically segmented.
2. Jointed appendages modified for various purposes.
3. Exoskeleton of chitin
4. Body usually divided into 3 regions, head, thorax and abdomen.
5. Possession of striated and non-striated muscles
6. Complete digestive system with mouth-parts for different methods of feeding
7. Open blood system, haemocyanin which are blue in colour
8. No nephridia
9. No cilia except in peripatus
10. Respiration is through body surface, spiracles, trachea, gills or lung books
11. Excretory system by green or antennary gland in aquatic ones and malpighian tubules in terrestrial ones.
12. Well developed nervous system
13. Sensory organs well developed consisting of eyes, antennae and antennules, balancing organs and in some auditory organs
Pronounced cephalisation with definite anterior end where sensory organs are concentrated.
14. Sexes nearly always separate
15. Some show metamorphosis

❖ The Arthropod Exoskeleton

- Epidermis secretes an external skeleton called the **exoskeleton**
- Advantages of possessing an exoskeleton:
 - Provides strong support
 - Provides rigid levers that muscles can attach to and pull against
 - Offers protection
 - Serves as a barrier to prevent internal tissues from drying out; important because many arthropods live on land
 - Serves as a barrier to prevent infection

❖ Molting

- In order to grow the arthropod must shed its exoskeleton, and secrete a new and larger one - **MOLTING** or **ECDYSIS**.



❖ Jointed Appendages

- Exoskeleton divided into a number of plates and cylinders



- At the junction point between plates and cylinders, the exoskeleton remains thin and flexible; these are the *joints*
- Jointed appendages allows arthropods to move efficiently and quickly
- Muscles are integral to arthropod movement; they attach to the inner side of the exoskeleton; they often function as a lever system

Specialized Arthropod Segments: Reduction in Metamerism

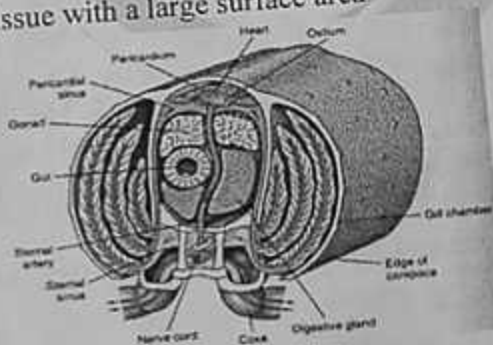
- The evolution of the arthropods witnessed a reduction in metamerism
- The arthropods evolved modified groups of segments (e.g., segments became lost, some fused together)
- The fusion of groups of segments into functional groups is called **tagmatization**
- In so doing, various appendages on segments became specialized for functions other than locomotion, e.g. prey capture, filter feeding, sensing various kinds of stimuli, gas exchange, copulation, etc.

❖ Arthropod Respiratory Advances

- Special respiratory structures allow the arthropods to metabolize more efficiently and thus move rapidly
- High metabolic rates require rapid oxygen delivery, and arthropods can accomplish this with respiratory organs that have a large surface area for collecting oxygen quickly

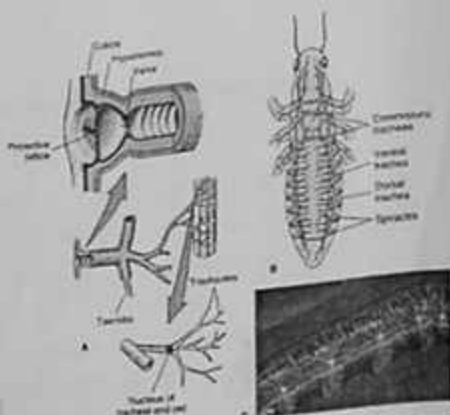
❖ Gills

- Many aquatic arthropods (crabs and lobsters) have **gills**, which are typically modifications of appendages or outgrowths of the body wall
- folds of tissue with a large surface area



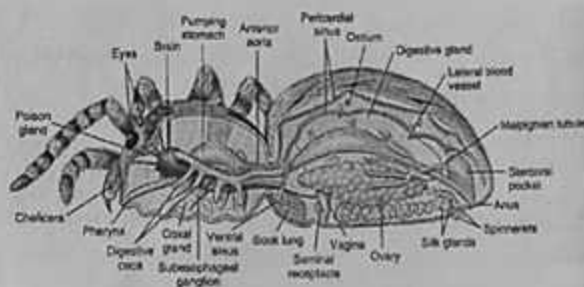
❖ Tracheae

- Gas exchange organs among terrestrial arthropods is usually internal; invaginations of the integument
- Insects have **tracheae**, branching networks of hollow air conducting tubes such that air is sent to every cell in every tissue



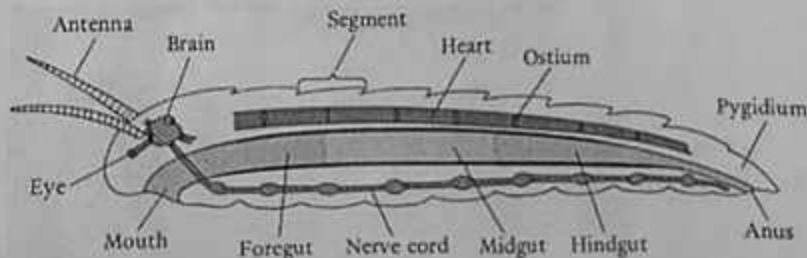
❖ Book Lungs

- Spiders have **book lungs**, chambers with leaf-like plates for exchanging gases; air flows over the plates and blood flows through them



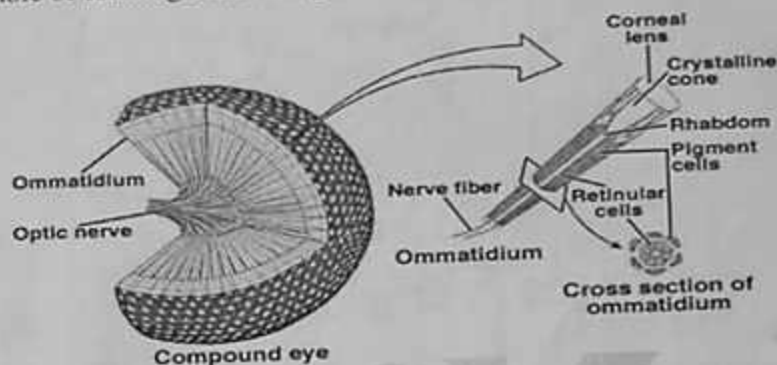
❖ Acute Senses

- Arthropods have a well-developed nervous system that is of the same overall design as the annelids; anterior brain and a double, ventral hollow nerve cord.
- The sensory receptors of arthropods are usually associated with modifications of the chitinous exoskeleton
- The head usually bears various kinds of sense organs (e.g. **antennae**) with extreme sensitivity



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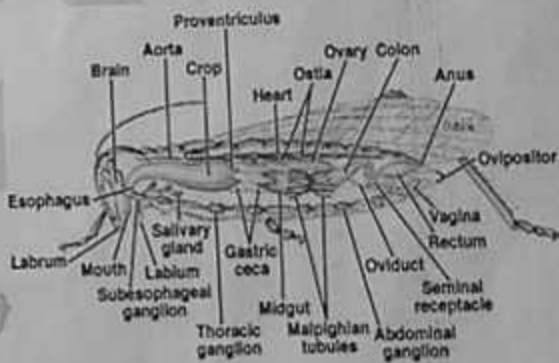
- Many arthropods have **compound eyes** - eyes that are composed of many visual units called **facets (ommatidia)**; capable of color vision and detecting the slightest movements of prey or predators
- Some eyes are simple eyes with only a few photoreceptors; however, they are capable of forming crude images



❖ Digestive System

Divided into 3 main regions: foregut, midgut, and hindgut

- Foregut and the hindgut are lined with chitin
- Foregut is involved with ingestion, mechanical breakdown, and storage
- Hindgut is involved with water absorption and formation of the feces
- Midgut is not lined with chitin; involved with digestion and absorption
- Outpockets (e.g. digestive glands) increase the surface area for digestion and absorption



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❖ Internal Transport and Excretion

- Open circulatory system
- Many crustaceans possess an excretory organ called the green gland (antennal gland), which filters fluid from the blood

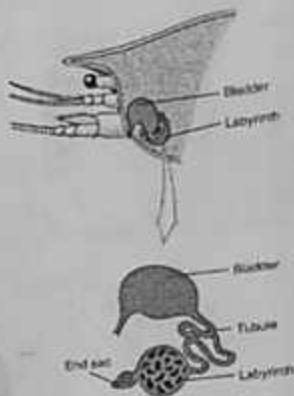
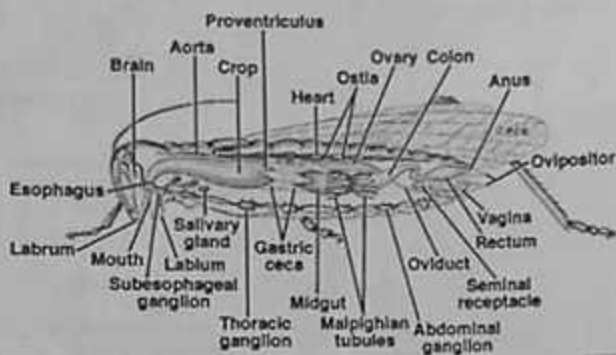


Figure 19-7
Schematic of antennal gland (green gland) of crustaceans. In various positions organs are much folded. Some crustaceans lack a labyrinth, and the secretory tubule (green gland) is a much coiled tube.

- Most insects and spiders have an excretory system called malpighian tubules

❖ Reproduction

- Sexes are separate; fertilization is external in aquatic forms, internal among the terrestrial forms

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ARTHROPOD DIVERSITY

❖ Subphylum Trilobita

- **Extinct Arthropod**
- A group of extinct marine arthropods.
- Ranged in size from a few millimeters up to 75 centimeters.
- Body divided into three parts:
 - **cephalon** (head), a single plate made up of several fused segments
 - **thorax**, consists of a number of segments hinged together
 - **pygidium** (tail), also segmented, but like the head, fused together into a single plate.
- **Compound eye**, composed of radially arranged visual units



Phylum Arthropoda

Extant Subphyla:

- Subphylum Myriapoda:
 - Class Diploda
 - Class Chilopoda
- Subphylum Crustacea:
 - Class Remipedia
 - Class Cephalocarida
 - Class Branchiopoda
 - Class Malacostraca (Orders Decapoda and Isopoda)
 - Class Maxillopoda
- Subphylum Hexapoda:
 - Class Insecta
- Subphylum Chelicerata:
 - Class Arachnida (Orders Scorpions, Acari, and Araneae)
 - Class Merostomata
 - Class Pycnogonida

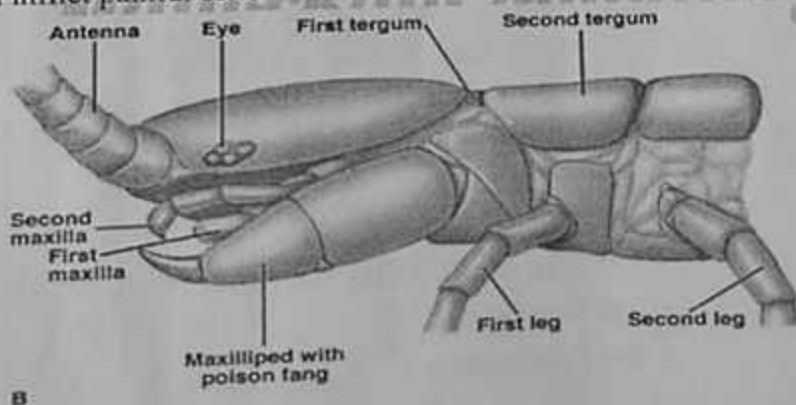
Overview Phylum Arthropoda

- 2 minor classes:
 - Class Merostomata- Horseshoe crabs
 - Class Pycnogonida- Sea spiders
- 5 major classes:
 - Chilopoda – centipedes
 - Diplopoda – millipedes
 - Malacotraca – crabs, lobster, crayfish, shrimp
 - Insecta – insects
 - Arachnida – spiders, scorpions, mites, chiggers

Subphylum Myriapoda

Class Chilopoda

- Centipedes
- 100 leggers
- Flattened worm like
- 1 pair of leg/segment
- Carnivores
- Possess a pair of poison claws or legs just behind the head which are used to paralyze their prey
- Feed on insects or small animals
- Most centipedes are beneficial
- Large species found in the tropics (length of up to 18 inches)
- Can inflict painful bites



- 1 pair of leg/segment

Subphylum Myriapoda
Class Chilopoda

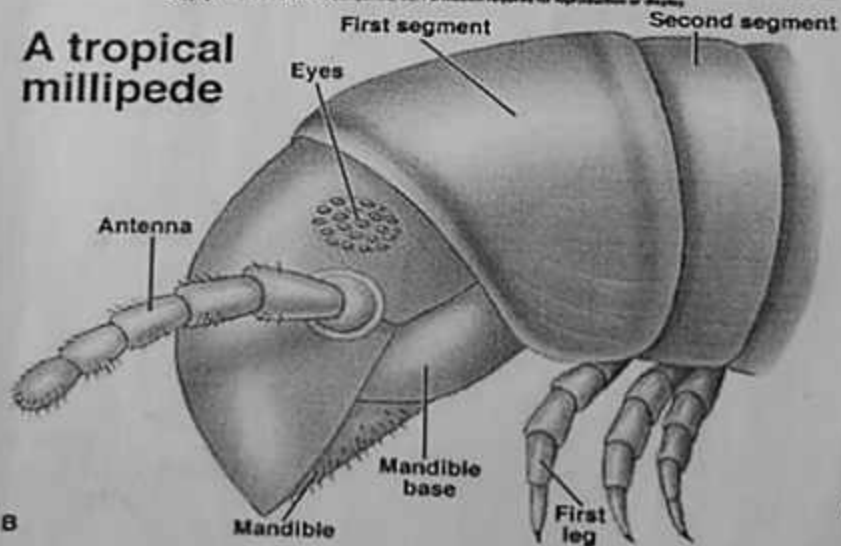


Subphylum Myriapoda
Class Diplopoda

- Millipedes
- 1000 leggers
- 2 pairs legs/segment
- Cylindrical worm like with 2/prs legs/seg
- Decomposers or herbivores
- >1000 species in US
- Short antenna
- Non-toxic
- 2 pair's legs/segment

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A tropical millipede



B

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**Subphylum Myriapoda-
Class Diplopoda**



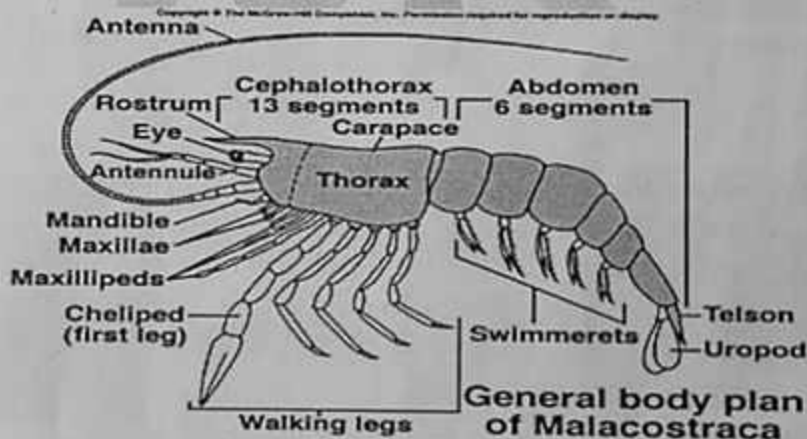
❖ Subphylum Myriapoda- Millipedes and Centipedes

❖ Life Cycle

- Millipede eggs are deposited in masses in the soil.
- Young millipedes have fewer segments than the adult, but they add segments at each molt of the outer shell, or exoskeleton.
- Molting occurs 7-10 times before maturity.
- Young millipedes mature the second year after hatching.
- Some species may live for several years.
- Similarly, centipedes molt several times, adding legs and body segments with each molt

❖ Subphylum Crustacea

- Body segmented with hardened Limbs
- Limbs generally with two branches
- 5 pairs of legs
- Two pairs of antennae
- Two compound eyes (eyes with many lenses)
- Body with 7 or more pairs of sometimes very different appendages for feeding, locomotion and sex
- Respiration by gills
- Nauplius- Distinctive larval stage with three pairs of branched appendages
- Although originally aquatic, many crustaceans are adapted to life on land
- The head bears a pair of compound eyes and 3 pairs of mouthparts: a pair of **mandibles**, and 2 pairs of **maxillae**; used for food handling
- Trunk varies considerably among classes
- Primitively, the first three pairs of thoracic segments are **maxillipeds**; function in handling food
- Also, there are usually 5 pairs of appendages strengthened for walking (walking legs) and protection (**chelipeds**, pincer-like claws)



❖ Examples of Crustaceans

- Shrimp
- Lobster
- Krill
- Pill bug
- Crabs
- Crayfish
- Barnacles (sessile)

Crustacean Diversity



Copepod

Ostracod



Water flea



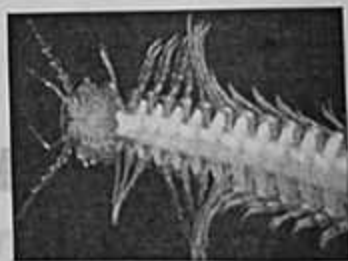
Fairy shrimp



Fish louse

❖ Subphylum Crustacea- Class Remipedia

- Cave-dwelling
- 30 segments with uniform, biramous appendages
- Caribbean, Indian Ocean, Canary Islands, Australia

❖ Subphylum Crustacea-
Class Cephalocarida

- Small and marine
- Uniform, leaf like triramous appendages



❖ Subphylum Crustacea-

Class Branchiopoda

- Freshwater mostly
- Leaflike appendages for respiration, filter feeding and locomotion
- Fairy, brine and clam shrimp and water fleas



❖ Subphylum Crustacea-

Class Maxillopoda

- Five head, six thoracic, and four abdominal somites with a telson
- Various fused segments
- No appendages on reduced abdomen
- Barnacles and copepods



❖ Subphylum Crustacea-

Class Malacostraca

- Modified appendages for crawling, feeding or swimming
- Lobsters, crayfish, crabs, shrimp, and isopods



Marine Maine Lobster



Crayfish

Crabs



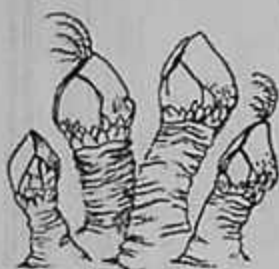
Male



Female



Barnacles



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❖ Subphylum Hexapoda-

Class Insecta

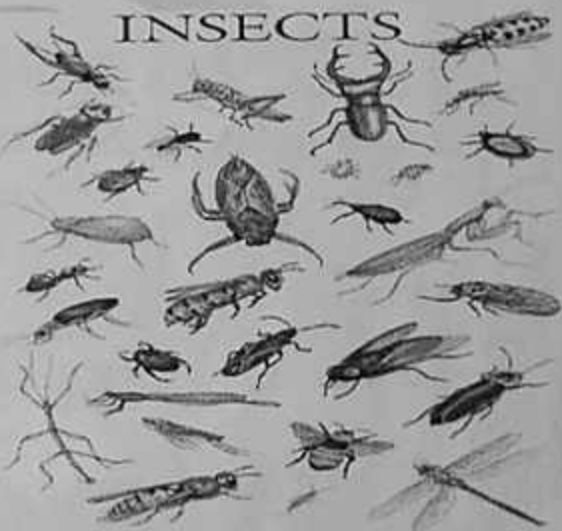
- >750,000 species
- Upwards of 3 million species
- Outnumber all other plant and animal groups
- Found in all environments (Social insect)
- 3 pairs of legs
- 10,000 ft. high
- Some fly 100s of KM per week

❖ Subphylum Hexapoda-

Class Insecta

- Three body divisions
 - Head
 - Thorax
 - Abdomen
- Exoskeleton
- 12 Inches—Largest
- 1MM—Smallest
- Rapid growth rate
 - Exponential
- Most mature < 1 year
- Display almost every color

INSECTS

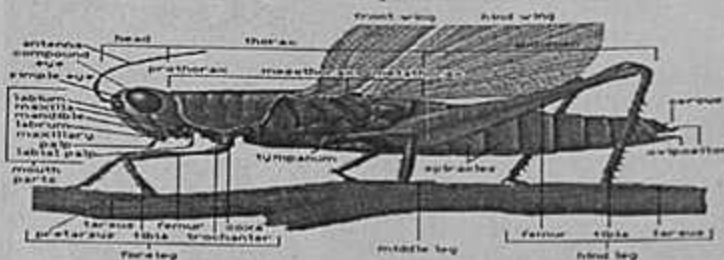


❖ Characteristics of Class Insecta

- Open circulatory system
- Spiracles along abdomen used for breathing
- Highly modified mouth parts (mandibles)
 - Some beetles can chew through lead or zinc
- Well developed tube within a tube digestive system
- Well developed nervous system
- Sight- two types of eyes
- Smell/taste-Use of antennae and feet
- Hearing- tympanum (eardrum)

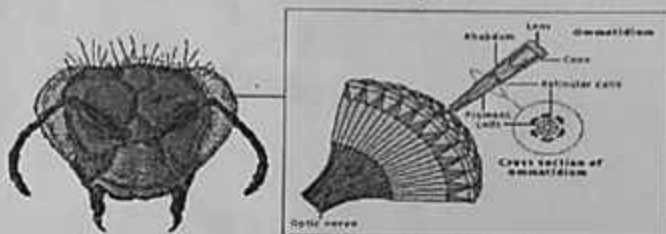
❖ Insect Body Plan

- Insects have 6 legs (3 pairs)
- 3 body parts
 - Head
 - Thorax
 - Abdomen
- Most insects have wings, however in many species these are vestigial
- Have advanced excretory system composed of **MALPHYGIAN TUBULES**
- Exchange gasses through a complex **TRACHEAL SYSTEM**



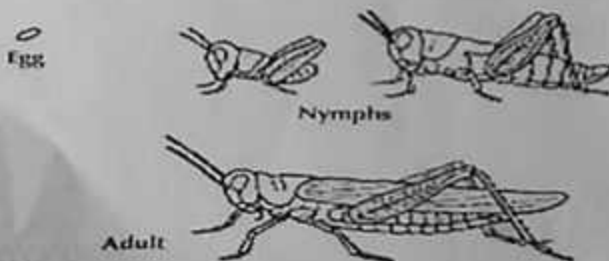
❖ Vision

- Have complex compound eye which is usually extremely sensitive to motion and allows 360° vision
- Most insects see well into the UV spectrum



❖ Metamorphosis

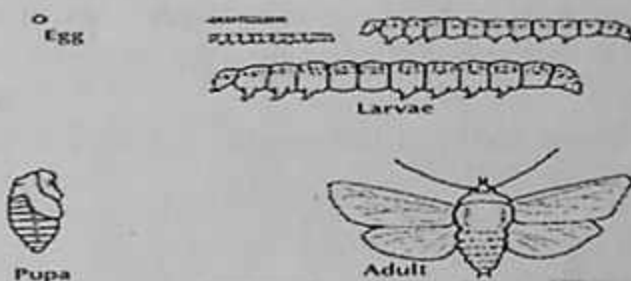
- Most insects undergo a process of metamorphosis - 2 types
- Incomplete metamorphosis
 - Larva similar to adult, with differing body proportions
 - Undergoes a series of molts resulting in adult phenotype

Incomplete Metamorphosis-Grasshopper

❖ Complete Metamorphosis

- Larva is very unlike adult phenotype
- Envelopes self in a cocoon or chrysalis where body breaks-down and reforms into adult form.

Complete Metamorphosis-Moth



❖ Subphylum Hexapoda-

Class Insecta

- Arachnology - study of spiders.
- Entomology - Study of insects and other arthropods
- Helpful or harmful
- Helpful insects:
 - Pollinators
- Research and discovery- fruit fly for genetic research
- Food producers- honey
- Textiles- Silk & beeswax
- Biological control agents

• INSECT HAVE ABOUT 29 ORDERS!!!!

❖ REASONS WHY INSECTS ARE THE MOST SUCCESSFUL AND ABUNDANT

1. Small size
2. Produce large number of eggs
3. Exoskeleton for conservation of water
4. Possession of different types of mouthparts for chewing, piercing, sucker create less competition among the insects
5. Excretory product of uric acid, so they lose little or no water
6. Leg appendages modified for jumping, leaping and swimming. Locomotion.
7. A pair of large compound eyes for easy sight
8. Protective colouration or a mode of protection against enemies

9. Ability to live anywhere, air, land, water.
10. Possession of trachea for respiration
11. Possession of antennae for feeling
12. Possession of wings for flight, escape and colonization of the species and finding of food
13. Ability to metamorphosise the metamorphosis stages can exist on their own because of their possession of adaptations for their own survival e.g. gills and spiracles
14. Colonies, social insect, ability to live together and work as a committee with division of labour.

❖ USEFULNESS OF INSECTS

1. Agents of pollination
2. Source of protein, source of food
3. Production of honey
4. Production of silk
5. Some enrich the soil
6. Some used as a means of biological control.

❖ Harmful effects

- Vectors of parasites, causing diseases - tsetse fly, mosquitoes
- Pests of fresh, stored products
- Some secrete poisons



Aphids

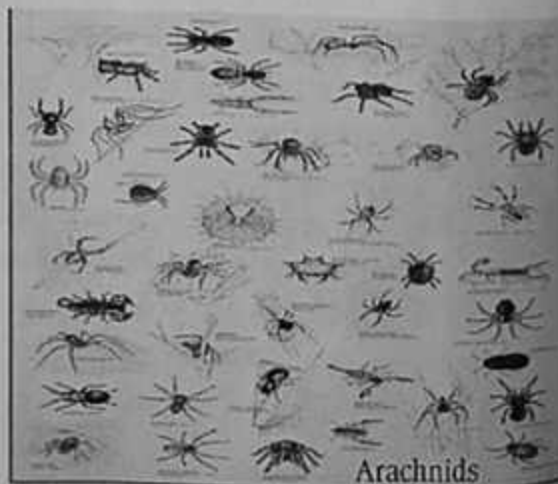


Aphids

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❖ Subphylum Chelicerata- Class Arachnida

- Spiders
- Mites/ticks
- Scorpions

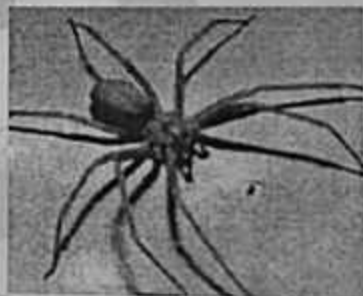


Arachnids

❖ Subphylum Chelicerata-
Class Arachnida
Order Araneae

- Most familiar spiders
- Cephalothorax (head fused to thorax) and abdomen
- All toxic
- No antennae
- Use pedipalps (2nd pair of appendages modified to catch and handle prey) and walking legs as sensory organs
- >30,000 Species/3,000 in U.S.
- Many, but not all spiders web builders
- Spinneretes- Appendages that secrete silk)
- Liquid feeders
- Inject digestive enzymes into prey and drink juices
- Use powerful jaws to crush exoskeleton- then digest and eat

Subphylum Chelicerata- Class Arachnida
Order – Araneae



**Subphylum Chelicerata-
Class Arachnida**

➤ **Order Scorpions**

- Tropical and subtropical
- Stinger on end of tail
- Pedipalps are pinchers
- Nocturnal
- Can be deadly but many aren't
- One of the oldest forms of life on Earth



❖ **Other Arachnids**



From left, engorged female,
unengorged female, male tick
as compared to a pin

• **Order Acari**

- Ticks, mites, and chiggers
- Found everywhere
- Omnivorous/scavengers/parasites
- Many blood feeders
- Disease vectors
 - Rocky Mountain Spotted Fever
 - Lyme Disease



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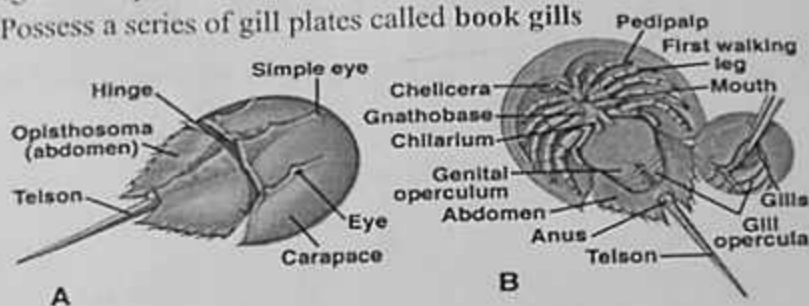
❖ **Other Chelicerates**

Class Meristomata (e.g., Horseshoe crabs)

- Marine chelicerates, common off of the Atlantic and Gulf coasts
- Large dorsal carapace bearing compound eyes
- Possess chelicerae, pedipalps, and 4 pair of walking legs (all but the last pair are chelate)

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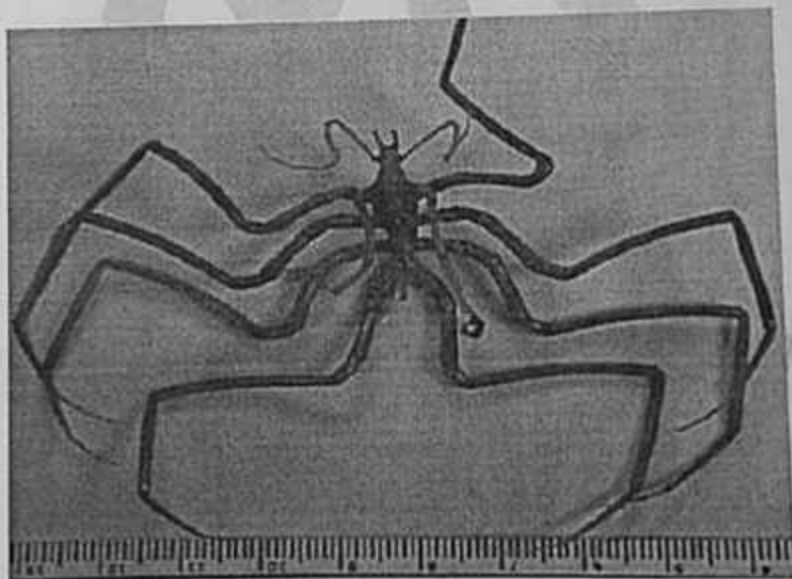
- Abdomen terminates in a long tail called the **telson**; used to turn the animal right side up
- Possess a series of gill plates called **book gills**

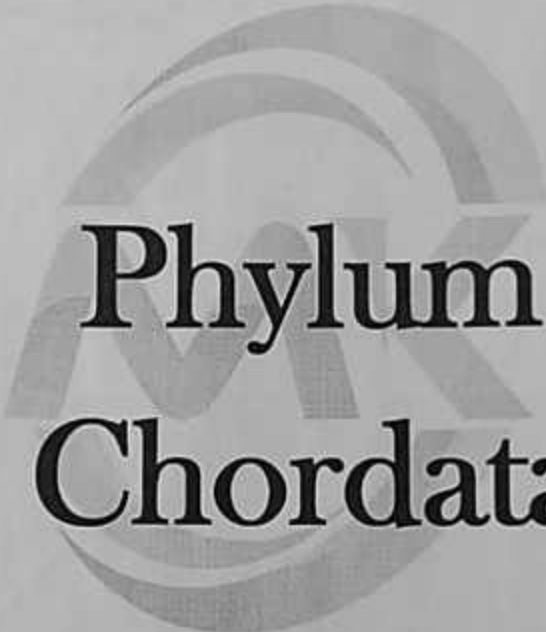


❖ Class Pycnogonida- Sea spiders

- Reduced abdomen
- No special respiratory/excretory organs
- Four to six pairs of legs

❖ Class Pycnogonida





Phylum Chordata

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MOGAJI, H.O.

Department of Animal and Environmental Biology

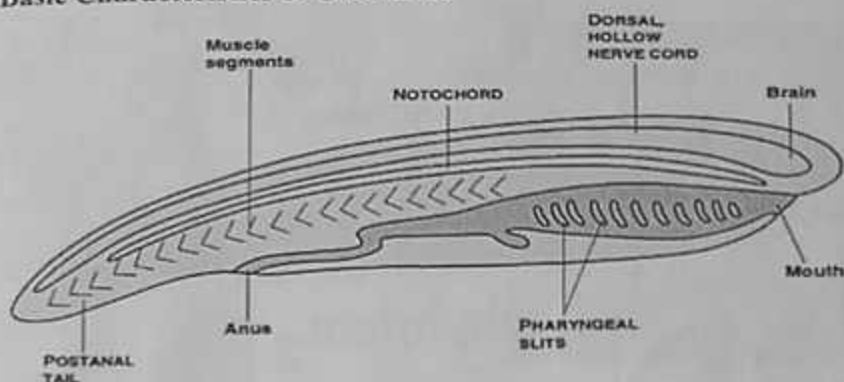
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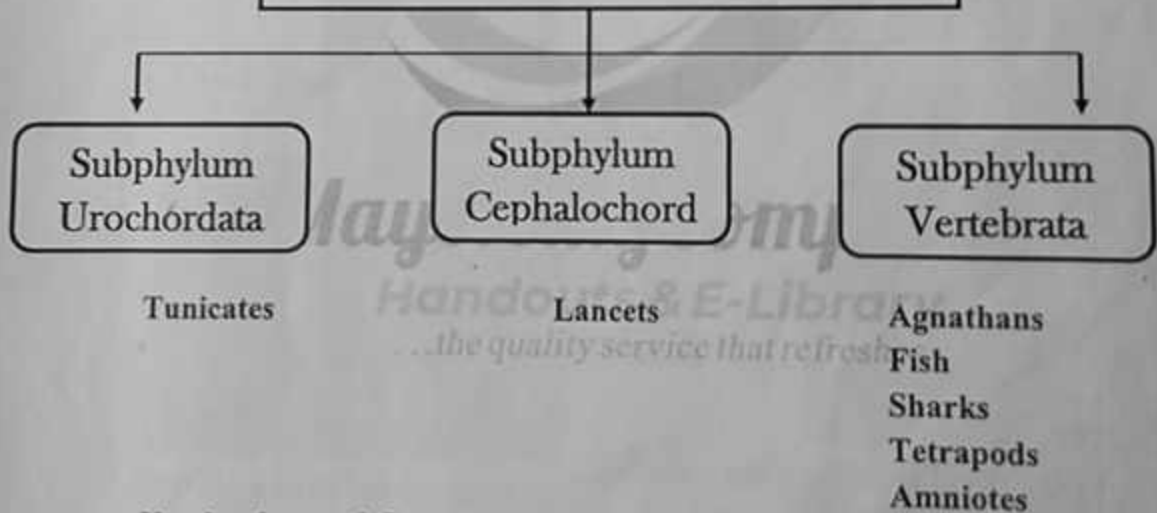
❖ **PHYLUM CHORDATA**

Animals that have a notochord

Basic Characteristics of Chordates



Phylum Chordata



*Urochordata and Cephalochordata are
Referred to as PROTOCHORDATES/ non-
vertebrate chordate*

❖ PHYLUM CHORDATA

Subphylum Urochordata - Tunicates

Class Ascidiacea - Sea Squirts

- Solitary e.g. *Styela montereyensis*
- Colonial - in groups but with own tunic
- Compound Colonial - colonial with shared tunic

Class Thaliacea - Salps (free swimming), planktonic

Class Larvacea - Gelatinous house, planktonic

Subphylum Urochordata

Tunicate



❖ Subphylum Urochordata = tunicate

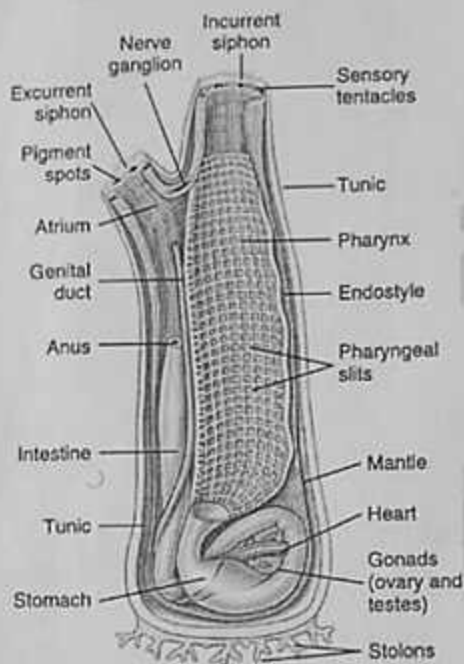
Tunicate - also called 'sea squirt'

- Notochord is confined to the tail
- Notochord is lost during metamorphosis into sessile adult
- Possess pharyngeal slits

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➤ Hermaphroditic & budding

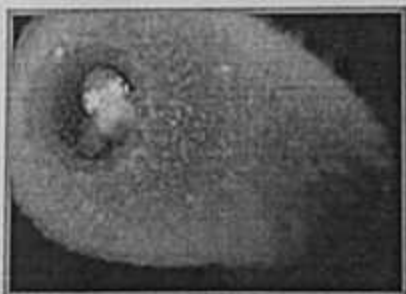
Tunicate Adult Anatomy



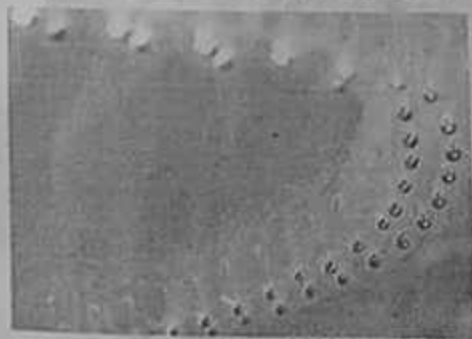
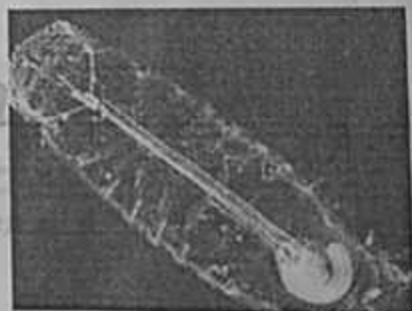
Class Ascidiacea - Sea Squirts



➤ Class Thaliacea - Salps (free swimming), planktonic



***Pyrosoma*- bioluminescent
Colonial salp**



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Class Larvacea - planktonic



Oikopleura



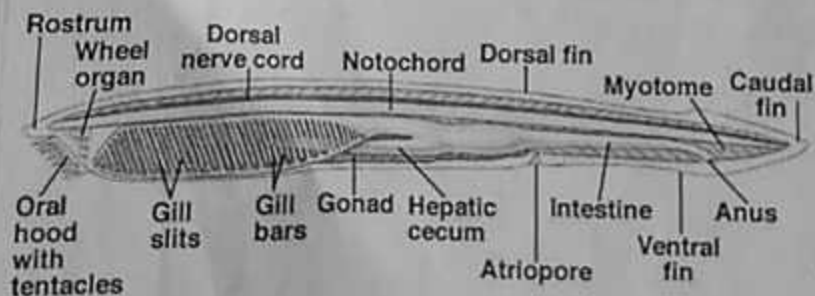
❖ Subphylum Cephalochordata

Lancet



- Strictly marine and transparent
- Live buried in sand with head sticking out
- Filter feeders

❖ Lancelet Anatomy



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❖ Vertebrate Higher Classification

Superclass Agnatha - without jaws

- Class Myxini – hagfishes
- Class Cephalaspidomorphi – lampreys

❖ Superclass Gnathostomata – with jaws

- Class Pisces
 - Chondrichthyes – sharks, rays, chimaeras
 - Osteichthyes – Bony fishes
- Class Amphibia – frogs, salamanders
- Class Reptilia – snakes, lizards, crocodiles
- Class Aves - birds
- Class Mammalia - mammals

❖ Agnatha (jawless fishes)

Lack:

- paired fins,
- scales,
- & well developed vertebrae

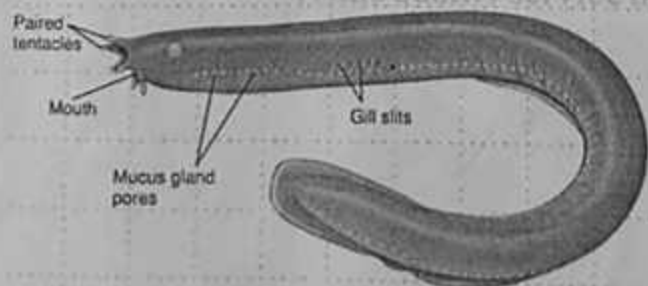
❖ Hagfish (slime eels)

- Mucus for protection
- Feed on decaying flesh

❖ Lampreys

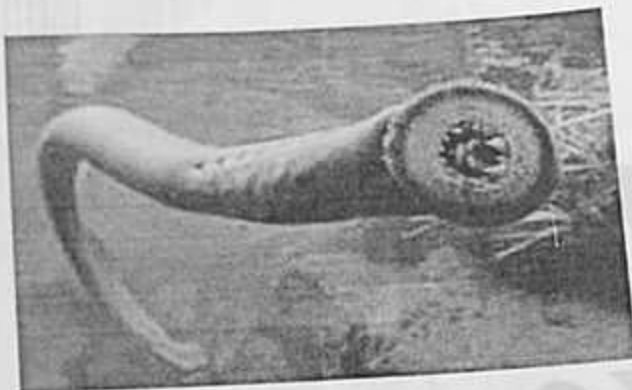
- Parasitic
- Anadromous

Marine adults, breed in freshwater



HAG FISH

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LAMPREY



➤ **PISCES (Fishes)**

These are the first vertebrate to make their appearance
All other vertebrates evolved from them

- Aquatic
- Ectothermic
- Scales
- Gills for oxygen exchange
- Lay eggs
- 2 chambered heart
- Lateral line system
- Pectoral, Pelvic other fins

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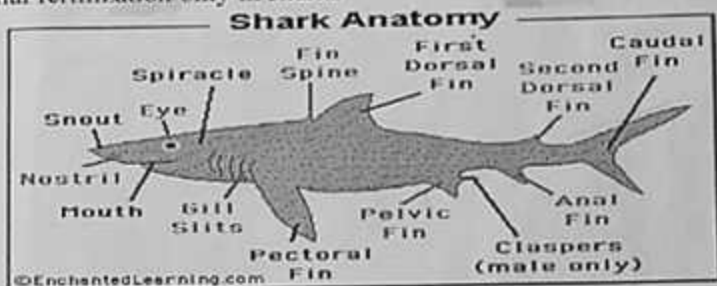
PISCES (Fishes)

49

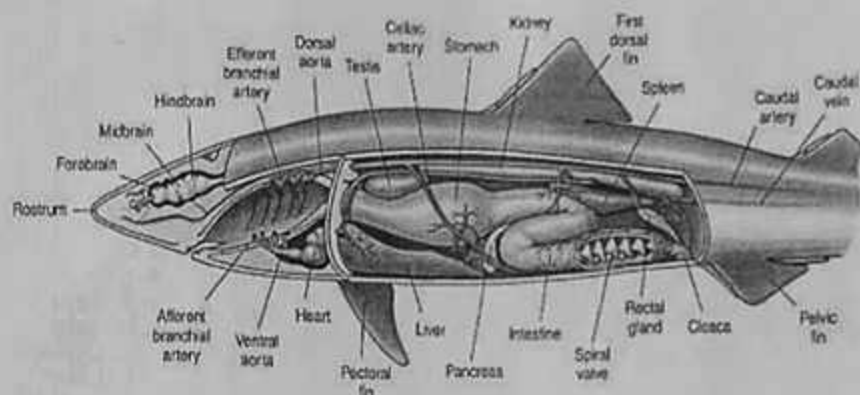
Chondrichthyes
(cartilaginous fishes)

Osteichthyes
(bony fishes)

- Chondrichthyes (cartilaginous fishes)
 - Cartilage endoskeleton
 - Heterocercal tail
 - Internal fertilization only in sharks

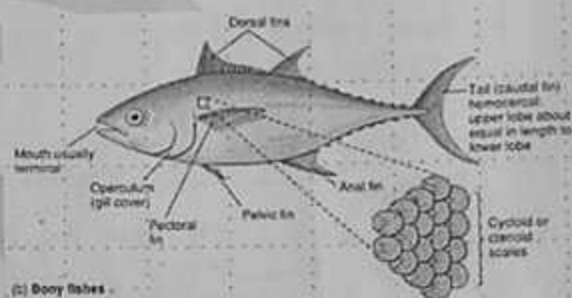


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❖ Osteichthyes (bony fishes)

- Swim bladder
- Operculum (gill cover)
- Homocercal tail
- Scales of bony origin
 - Smooth cycloid
 - Spiny ctenoid



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EVOLUTIONAL ADVANCEMENT

PISCES, AMPHIBIANS, REPTILES, BIRDS,
MAMMALS

Tetrapods

Tetrapods: Having 2 pair of limbs

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*Aquatic vertebrates wanted to colonize the land, so
they developed limbs*

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❖ **Class Amphibia**

➤ Amphi = Dual ; Bios = Life

Amphibians = Double life

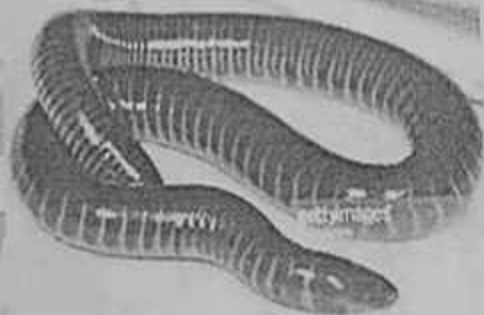
Amphibians are animals that live both on land and water

❖ **Characteristics**

- Cold blooded
- Returns to water to breed
- Metamorphosis
- Some toxic
- Estivation-dry and hot
- Hibernation-cold
- No scales/claws

3,500 species❖ **3 Major Groups**

- Apodan (No legs)
- Urodeles (have legs and tail)
- Anurans (have legs, no tail)

APODANS**CAECILIAN****URODELES (CAUDATA)**

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siren
(Siren lacertina)

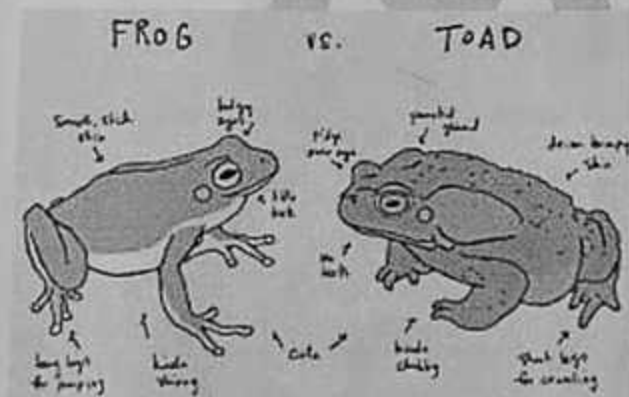
Salamander

- Newt
- Siren

alamy stock photo



ANURANS



- Frogs
- Toads

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❖ CLASS AMPHIBIA



Salamander



newt

Mudpuppy
(Salamander)

Poison arrow frog

*Rana cancrivora*

Coqui

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EVOLUTIONAL DEVELOPMENT

PISCES, AMPHIBIANS, REPTILES, BIRD, MAMMALS Amniotes

Amniotes: Eggs in which the embryo develop inside an amnion (Fluid)

Terrestrial vertebrates got tired of going back to water to breed, so they started laying amniotic eggs

❖ Class Reptilia

Reptilia form the latin word "Repto"

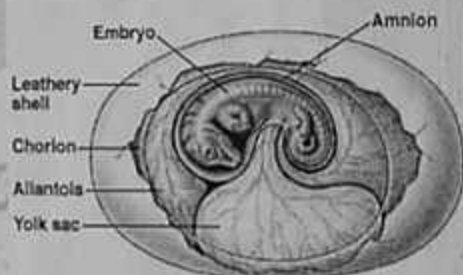
Repto = Crawl.

Reptiles are scaly terrestrial vertebrates that crawls

Amniotic Egg of Reptile

❖ Characteristics

- Cold blooded
- Have scales/claws
- Lay Amniotic egg
- Dry skin
- 3 chambered heart (except crocks)



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❖ 3 Major Groups

- Chelonia (hard Shell)
- Squamata (have scale)
- Crocodila

CHELONIA



Tortoise (Terrestrial)
Turtles (Aquatic)



Are you Ophidiophobia?...



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❖ SQUAMATA

Snelly

Lizards, House geckos, chameleon,
snakes, dragon e.t.c



❖ CROCODILA

Crocodiles and Alligators



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❖ Class Reptilia



Marine iguana



Saltwater crocodile



Marine turtle



Sea snake

❖ CLASS AVES

Aves = Birds

First vertebrate to colonize the air

Most successful vertebrates

❖ Characteristics

- Warm blooded
- Feathers and wings
- Hollow bones
- Horny bill
- Lungs have air sacs
- Hard-shelled amniotic egg

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❖ 2 Major Groups

- Paleognathae
(Flightless birds) - Ratites
- Neognathae
(Flying birds) - Carinates

Flightless Birds

Ostrich, Kiwi, Emu, Duck, e.t.c



KIWI

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❖ Carinates

Owl, Eagle, Vulture, e.t.c



❖ CLASS AVES



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❖ Class Mammalia

- Mammals from mamma = breast
- Animals that possess mammary glands.
- Most advanced group of vertebrates

❖ Characteristics

- Warm blooded
- Have fur or hair
- Suckle young
- 3 middle ear bones

❖ 3 Major Groups

- Protheria (monotremata)
 - Egg laying
- Metatheria (Marsupials)
 - Pouched mammal
- Eutheria (Placental mammals)
 - True mammals

❖ EGG LAYING MAMMALS

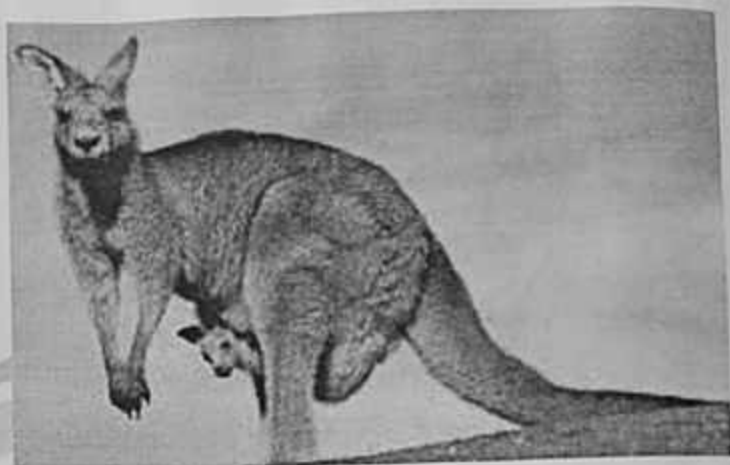
Duck billed and flat billed platypus



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❖ Pouched mammals

Kangaroo, Oposum, koalas



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❖ Placental mammals

Bats, whale, cats, dog,
dolphin, elephant, Lion e.t.c

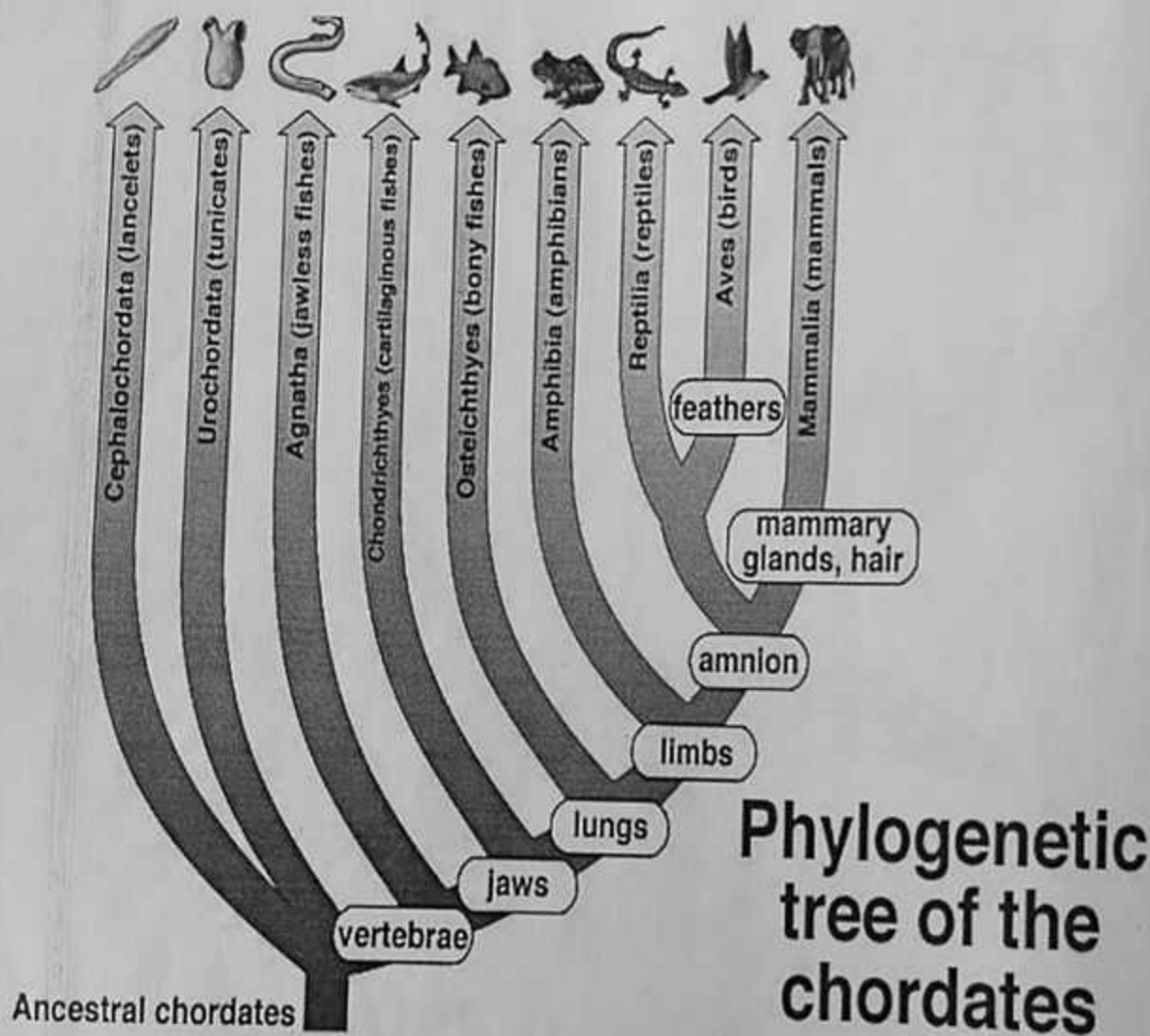


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Phylum Echinodermata

echinos = spiny derma = skin

ata = to bear

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❖ GENERAL CHARACTERISTICS

- They are un-segmented
- Coelomate animals
- Bilaterally symmetrical in the larval stage.
- Pentaradially symmetrical in the adult
- They possess a DERMAL SKELETON which consists of calcareous ossicles
- These bones may develop short or long spines.
- The nervous system is diffuse and remains attached to the epidermis.
- No nephridia or any other special excretory organs.
- The sexes are separate and gonads discharge by special ducts directly to the exterior.
- The echinoderms are very common as fossil forms (since the Cambrian period) and are entirely marine.
- Water-vascular system used in locomotion, attachment, &/or feeding.
- None of them are parasitic

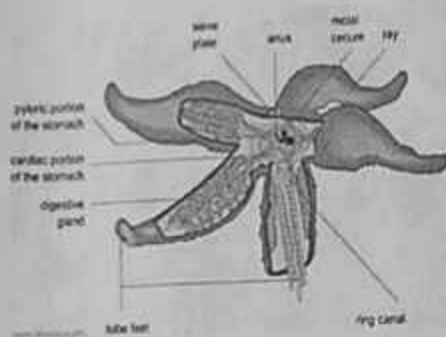
❖ Habitat:

- - Marine environments
 - Bottom of deep seas
 - Bottom of coastal shores

Niche:

- - Relatively slow moving
- - Feeding
 - Some species feed on animal remains on the ocean floor
 - Some filter plankton through their mouth pore
 - Some scrape food off rocks
 - Others are predaceous on mollusks, arthropods

❖ Digestive System:

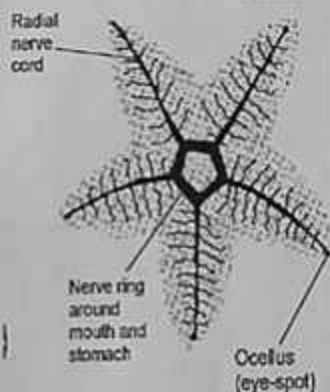


- Have an anus, but it is almost nonfunctional; undigested food is expelled back through the mouth
- Respond to light, chemicals, and various mechanical stimuli

❖ Reproduction:

- Diocious – the two sexes are indistinguishable externally
- Gamete release by one individual is accomplished by the release of spawning pheromones, which induce other sea stars in the area to spawn, increasing the likelihood of fertilization

❖ Nervous System:



- Nerve ring that encircles the mouth
- Radial nerves that extend into each arm (these coordinate the functions of the tube feet)
- Other nervous elements are in the form of a nerve net associated with the body wall

❖ The phylum Echinodermata is classified into two sub-phyla:

- Subphylum I Eleutherozoa
- Subphylum II Pelmatozoa

❖ Subphylum Eleutherozoa

- The Subphylum Eleutherozoa is free living forms, without a stalk.
- They have 4 living classes
 - Class I Asteroidea e.g. Starfishes
 - Class II Ophiuroidea e.g. Bristle Stars
 - Class III Echinoidea e.g. Sea urchins
 - Class IV Holothuroidea e.g. Sea cucumber

❖ Class Asteroidea

Examples: sea stars (starfish)

Sea stars:

1. Most common echinoderm; central disc and 5 arms (or rays) radiating from disc
2. May be various colors of red, purple, green, blue and yellow
3. Range in size from about 2 cm to nearly a meter
4. Found on pier pilings and rocks in tide pools along coasts

It has no brain

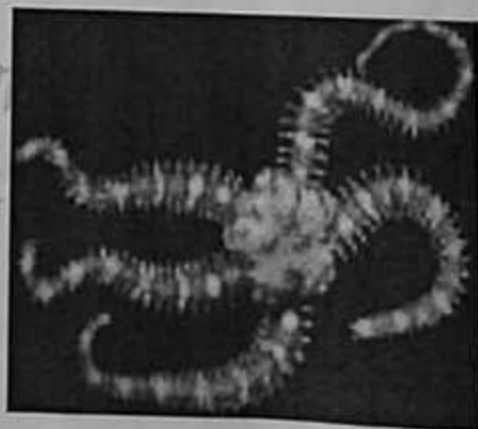


❖ Class Ophiuroidea

Examples: brittle stars

Brittle stars:

1. A distinct disc set apart from the arms
2. Slender, articulating arms
3. Rapid, serpentine (snakelike) movements
4. Found on the seashore, burrowed in sand or deep sea sediments or under rocks and kelp



❖ Class Echinoidea

Examples: sand dollars, sea urchins

Sand dollars



1. Range in size from 1-15 cm
2. Flattened skeleton (test) covered with a dense thicket of tiny spines
3. Aboral surface exhibits flower petal-shaped grooves (petaloids) that correspond to the arms of sea stars and brittle stars
4. Tube feet in the petaloids are used in locomotion

❖ Class Echinoidea

Examples: sand dollars, sea urchins



Sea urchins

1. Rounded body shape
2. Long spines on exterior
3. The areas corresponding to the rays of the sea stars are fused

❖ Class Holothuroidea

Examples: sea cucumbers, feather stars

Sea cucumbers

1. Long, cucumber like body lacking a solid, calcareous skeleton
2. Oral end has a ring of retractile tentacles that represent highly modified tube feet
3. Found on the sea bottom, often partially submerged in mud or sand, or among intertidal rocks



❖ Class Holothuroidea

Examples: sea cucumbers, feather stars

Feather stars

1. Most primitive of the living echinoderms
2. From a small cup or calyx, protrudes five flexible arms (rays) with branches (pinnules) very much like pine needles

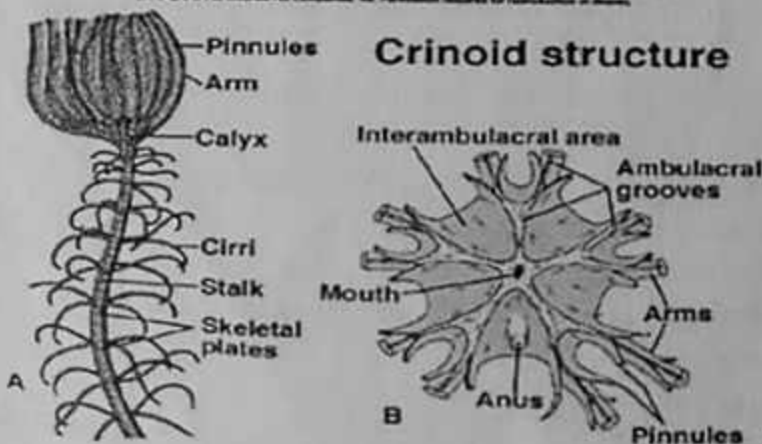


❖ Subphylum Pelmatozoa

- The subphylum Pelmatozoa are sessile forms, usually stalked,
- Represented among the living echinoderms by the class Crinoidea e.g. Sea lilies and Sea feathers.

❖ Class Crinoidea

- Most primitive of the echinoderms.
- Unusual in that the oral surface is directed upward
- Aboral surface is attached to the substrate by means of a bendable **stalk**
- The portion of the crinoid body attached to the stalk is called the **crown**; bears a number of arms
- Along the length of the arms are branches called **pinnules**
- The arms and the pinnules have ambulacral grooves with suckerless podia (secrete mucus)
- The ambulacral grooves are heavily ciliated and the cilia is used to direct food to the mouth (=filter feeding)



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Phylum Nematoda

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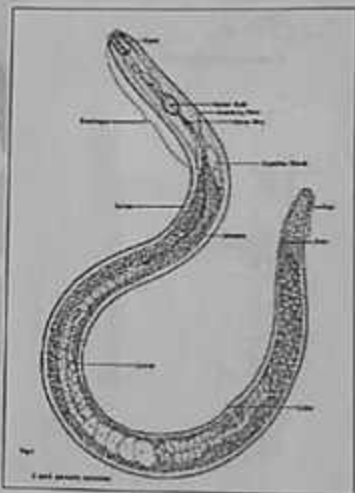
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❖ Phylum Nematoda: Diversity

- Free-living forms found in nearly every environment i.e they are cosmopolitan
 - Free-living marine & freshwater
 - Between grains of beach sand
 - Key soil dwellers (nutrient processing)
 - Polar ice fields
- Key plant & animal parasites
- Triploblastic and bilaterally symmetrical

❖ Phylum Nematoda: General Biology

1. Habit and Habitat
2. External Features
3. Body Wall
4. Body Cavity
5. Digestive System
6. Respiration
7. Excretory system
8. Nervous system
9. Sense organs
10. Reproductive system



❖ HABIT AND HABITAT

- Cosmopolitan in distribution
- Freshwater, marine and soil dwellers
- Majority are free living, some are parasite in plants and animal
- Mode of Nutrition is Holozoic
- Most free living <2.5 mm in length. Some parasites > 50 cm in length.

❖ EXTERNAL FEATURES

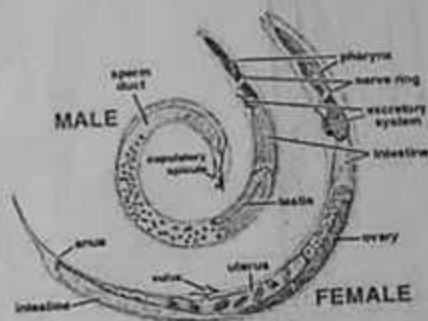
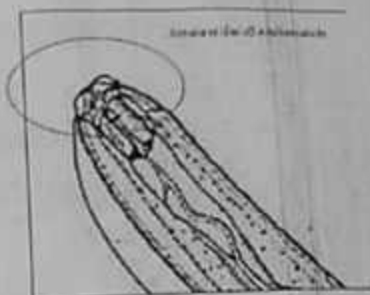
- Body covered by a tough, smooth and elastic cuticle.

Anteriorly,

- Possession of six lips (Labia) fused up in some,
- Presence of olfactory chemoreceptors called Amphids

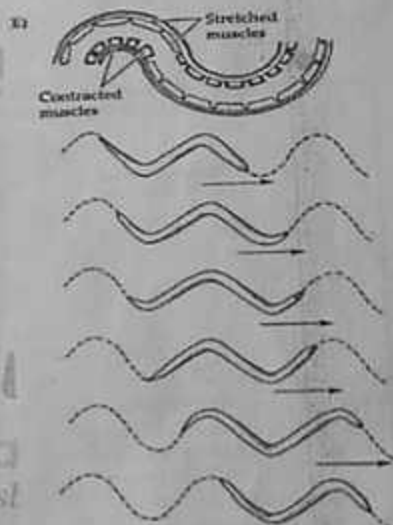
Posteriorly,

- Presence of anus with thick lips
- Male has cloaca from which two equal Chitinous spicules (penial setae) projects.
- Presence of papillae in male connected with copulation.
- Presence of short post-anal tail Straight in female, curled in male
- Male smaller and thinner than females
- Presence of genital aperture (vulva or gonopore) in female on the ventral side.
- Presence of excretory pore at mid ventral location.



❖ Movement

- Muscles
 - Longitudinal muscles only
 - Several along body length
 - Must push against solid surface to move forward
 - Elasticity of cuticle may aid in movement
- Antagonistic muscle action
 - Right contracted → left relaxed

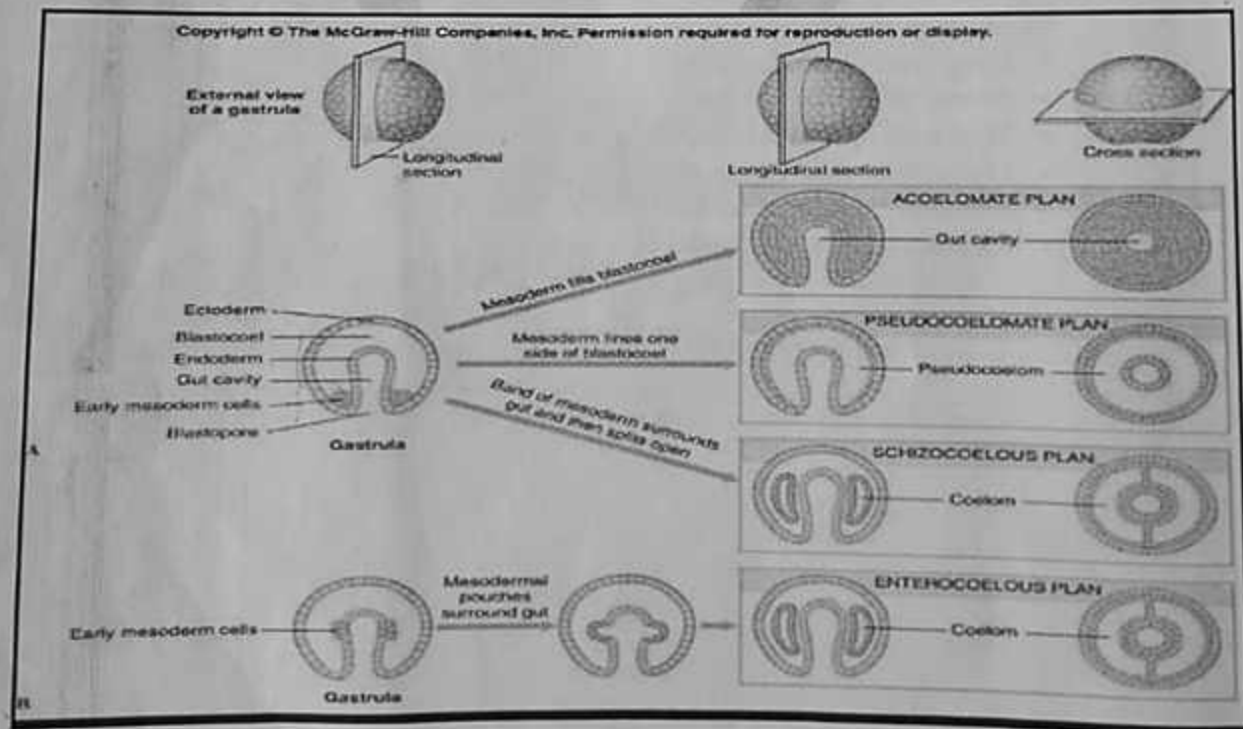
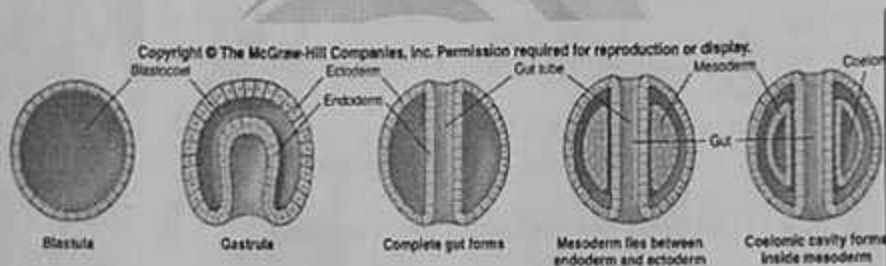


Illustration, Second 2

❖ BODY CAVITY

Nematodes are pseudocoelomate
i.e mesoderm lines up the blastocoel

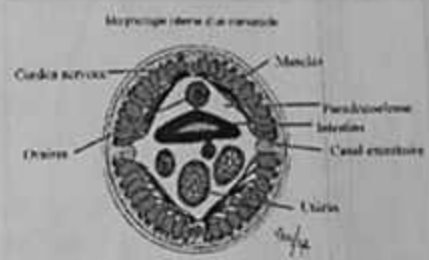
COELOM



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BODY CAVITY or PSEUDOCOEL

1. Pseudocoels have fibrous tissue and fixed cells called coelomocytes or giant cells.
2. The pseudocoel is filled with fluid called Pseudocoelomic fluid and perienteric fluid.
3. This fluid distributes digested fluids and various metabolites.
4. It is composed of 93% water and 7% solids like protein, glucose, sodium chloride, phosphate e.t.c



❖ BODY SYMMETRY

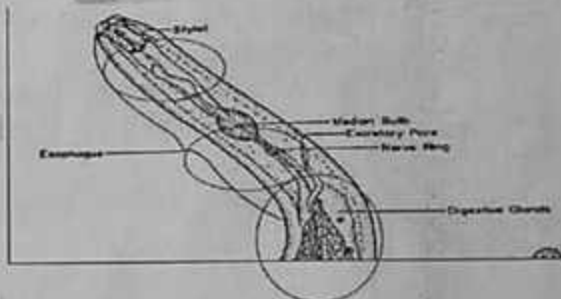
- Bilateral symmetry
- Triploblastic

❖ DIGESTIVE SYSTEM

- Consist of :
 - Mouth (lips, teeth, Stylet and jaw)
 - Short Muscular pharynx
 - Oesophagus

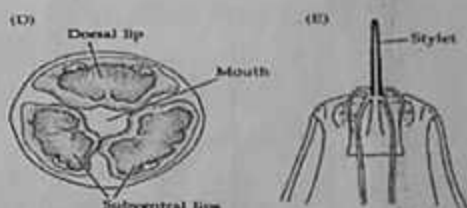
Oesophagus forms

- 1 Foregut; long intestine
- 2. Midgut
- 3. Hindgut: Short rectum



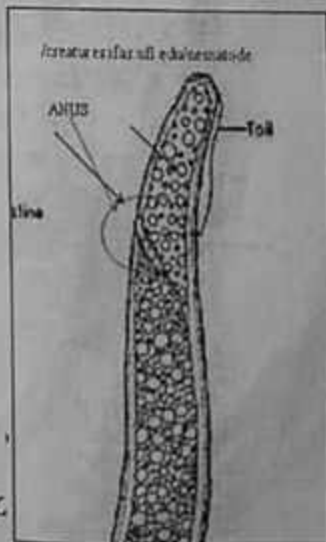
- Buccal Cavity varies depending on prey

- Bacteria - no teeth or stylet
- Plant root - stylet
- Carnivore - small teeth and sometimes stylet
- Intestine - large hook-like teeth



- Hindgut

- Hindgut opens to a cloaca in male, but in females open to an anus.
- Contraction causes fecal materials to be discharged.



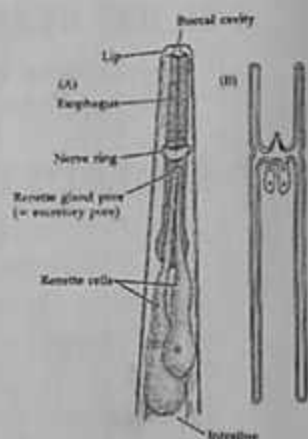
MAYORK

❖ FOOD AND FEEDING

- Foods comprising blood, tissue, bacteria, plants are partly or fully digested food of host.
- Food sucked in by suctional action of pharynx
- Digestion is extracellular in intestinal lumen
- Digestion facilitated by proteases, amylases and lipases secreted by glands of the pharynx
- Digested nutrients absorbed by microvilli on intestinal wall and distributed in pseudocoelomic fluid.
- Excess food stored as reserve glycogen in intestinal wall and muscles

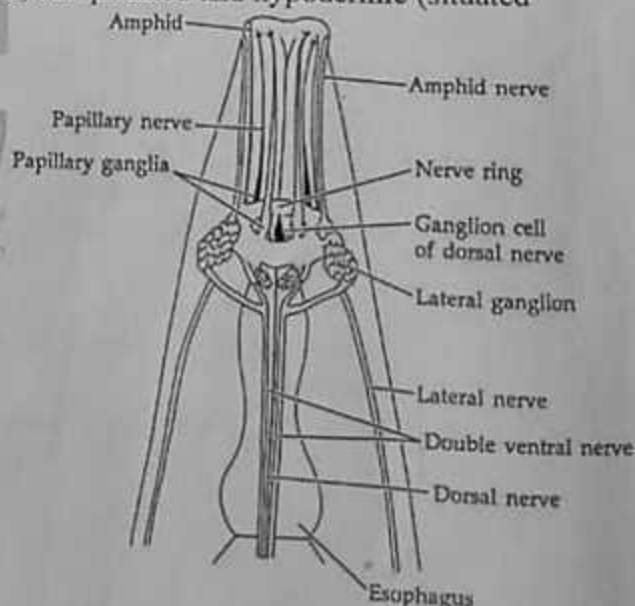
❖ EXCRETION

- Excretory system consists of one or two large RENETTE glands cells lying in the pseudocoel near the pharynx and intestine.
- A duct arise from each RENETTE gland cell and open by an excretory pore
- Excretion also via the digestive system
- Excretory system regulates, Water-salt balance, Ionic balance
- Body wall helps to excrete Ammonia



❖ Nervous system

- Nervous system is well developed, complicated and hypodermic (situated in the body wall)
- It is organized into:
 - Nerve ring
 - Plus associated ganglia
 - Sensory nerve input
- Four major nerve cords:
 - 1 dorsal,
 - 1 ventral,
 - 2 lateral



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❖ Sensory structures

S/N	Sensory Structure	Location	Function
1.	Labial Papillae	Anterior	Taste
2.	Amphids	Anterior	Olfactory chemoreceptors
3.	Phasmids	Anterior	Chemoreceptors
4.	Cephalic Papillae	Anterior	Chemoreceptors
5.	Cervical Papillae	Anterior	Tactile
6.	Genital Papillae	Posterior	Tactile and copulation

❖ RESPIRATORY SYSTEM

- RESPIRATORY ORGAN IS ABSENT.
- However parasites carry on ANAEROBIC respiration.
- They break down glycogen into Carbon dioxide and Fatty acids which are excreted through the cuticle.

❖ Reproduction

- Sexual reproduction
- Sexes are separate i.e. dioecious
 - Internal fertilization
 - Male has cuticular spicules
 - Males have curled end and are shorter
- Sperm lack flagella (amoeboid)
- Oviparous or ovoviviparous
- High incidence of parthenogenesis
 - In some, mating occurs but sperm nucleus not used.
- Some are hermaphrodites

❖ **Male Reproductive organs**

- Testis
- Vas deferens
- Seminal vesicle
- Ejaculatory duct
- Penial setae

❖ **Female Reproductive organs**

- Ovaries
- Oviducts
- Uteri
- Vagina

❖ **ECONOMIC IMPORTANCE**

- Nematodes are either free living or parasitic.
- Plant Parasite
- Animal Parasite
 - Veterinary importance (Livestock, ruminants e.t.c.)
 - Medical importance (humans)

❖ **PHYLUM NEMATODA: CLASSIFICATION**

- KINGDOM – Animalia
 - Classified based on the presence or absence of a caudal sense organ
- PHASMID**
1. Class Adenophorea (Aphasmidea -those without phasmids)
 2. Class Secernentea (Phasmidea -those with phasmids).

❖ **CLASSIFICATION OF PARASITIC NEMATODES**

- Soil Transmitted Nematodes
 - Infection usually through the faecal oral route
 - Exceptions, skin penetration
- Blood/Tissue dwelling Nematodes
 - Infection usually through bites of an arthropod
 - Exceptions, faecal oral route

❖ 17 Orders

1. Order Enploidea – *Enoplus*
2. Order Dorylaimoidea- *Dorylamius*
3. Order Mermithoidea- *Mermis*
4. Order Chromadoroidea – *Paracytholamius*
5. Order Araelaimoidea – *Plectus*
6. Order Monhysteroidea- *Cylindrolaimus*
7. Order Desmoscaleicoidea – *Tricoma*
8. Order Rhabditoidea- *Rhabditis*
9. Order Rhabdiasoidea- *Entomelas*
10. Order Oxyuroidea – *Oxyuris*
11. Order Ascaroidea- *Ascaris*
12. Order Strongyloidea – *Ancylostoma*
13. Order Spiruroidea – *Oxyuris*
14. Order Dracunculoidea – *Dracunculus*
15. Order Filarioidea – *Wuchereria*
16. Order Trichuroidea – *Trichuris*
17. Order Dioctyphymoidea - *Eustrongylides*

❖ *Ascaris Lumbricoides* (Common name: roundworm)

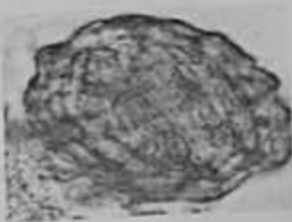
- Phylum Nematoda
- Class: Secernentea
- Order: Ascaroidea
- Family: Ascaridae
- Genus: *Ascaris*
- Species : *Ascaris lumbricoides*

Parasitic life cycle, medical importance

❖ *Ascaris lumbricoides* (Round worm)

- Adult worms live in the lumen of the small intestine.
- A female may produce approximately 200,000 eggs per day, which are passed with the feces
- After infective eggs are swallowed, the larvae hatch, invade the intestinal mucosa, and are carried to the lungs.
- The larvae penetrate the alveolar walls, ascend the bronchial tree to the throat, and are swallowed.
- Upon reaching the small intestine, they develop into adult worms

❖ Forms and Shapes of *A. lumbricoides*



❖ Forms and Shapes of *A. lumbricoides* cont'd

- Egg - characteristic of the genus. Size & shape are relatively consistent.
- Larvae - undergo several molts
 - Rhabditiform Larvae
 - Filariform Larva (third stage usually the infective stage)
- Adult - varies in size from genus to genus; Range from less than 1 mm to over one meter.

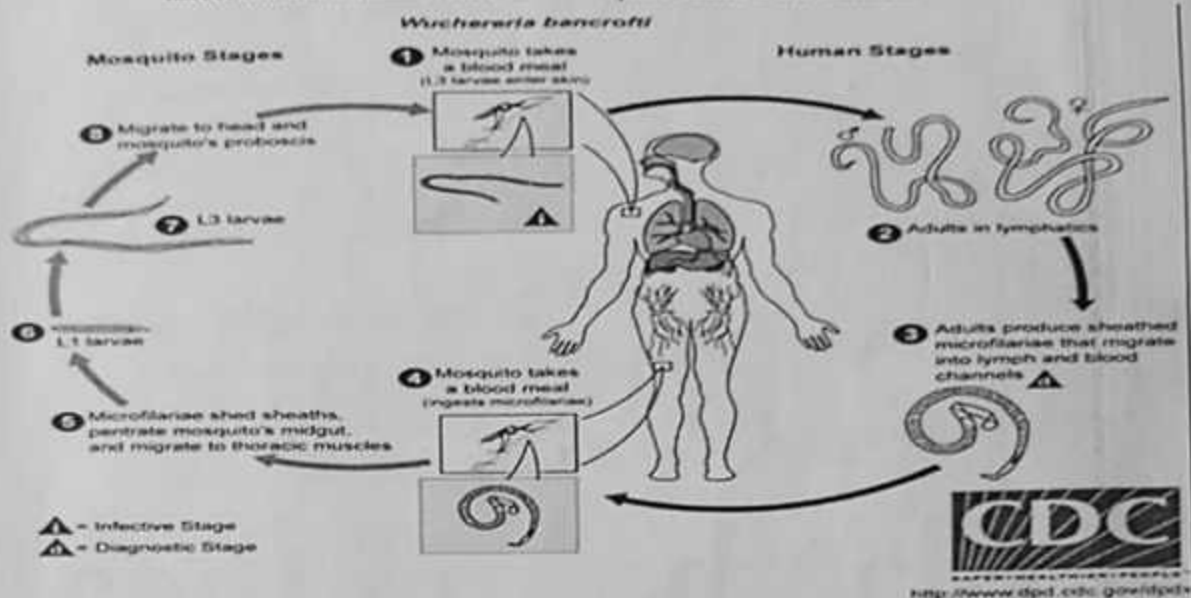
❖ *Wuchereria bancrofti* (Filariaworm)

- Kingdom: Animalia
- Phylum: Nematoda
- Class: Secernentea
- Order: Filarioridea
- Family: Onchocercidae
- Genus: *Wuchereria*
- Species: *Wuchereria bancrofti*

❖ *Wuchereria Bancrofti* (Life cycle)

- During a blood meal, an infected mosquito (*Anopheles* or *Culex* spp) introduces microfilarial larvae onto the skin of the human host, where they penetrate into the bite wound.
- They develop into adults that commonly reside in the lymphatics
- The microfilariae migrate into lymph and blood channels moving actively through lymph and blood.
- Another mosquito ingests the microfilariae during a blood meal and work their way to the mosquito's midgut and develop into infective larvae.

- The infective larvae migrate through to the mosquito's Prosbocis and can infect another human when the mosquito takes a blood meal.



❖ Pathology: hydrocele



Elephantiasis



READ MORE!!

- *Trichuris trichiura* (whip worm)
- *Ancylostoma duodenale* (hook worm)
- *Necator americanus* (hookworm)
- *Strongyloides stercoralis* (thread worm)
- *Trichinella spiralis*
- *Onchocerca volvulus* (Filaria worm)
- Their Biology and Classification?

RECOMMENDED TEXT

- INVERTEBRATE ZOOLOGY by E.L Jordan and Dr. P.S. Verma

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Phylum Platyhelminthes

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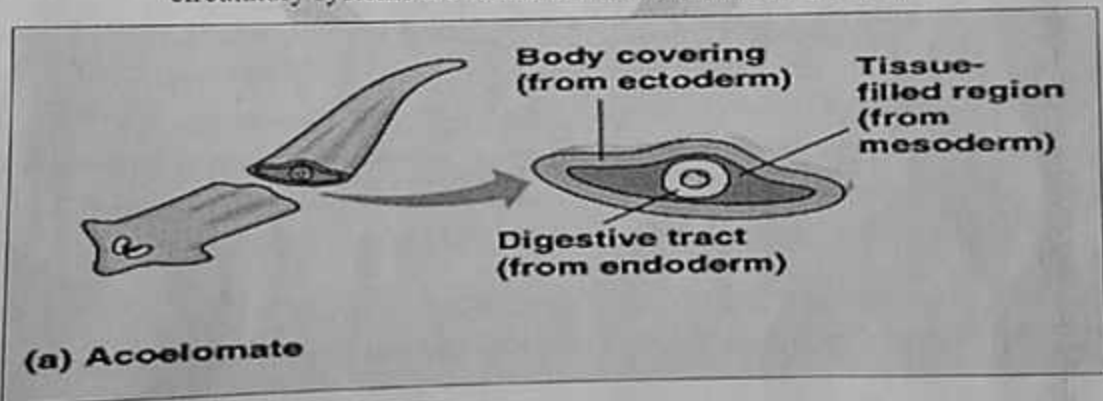
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❖ Worms

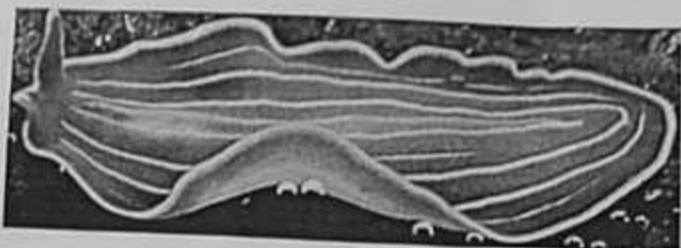
- The term worm is loosely employed in biology and is applied to very different animals.
 - Segmented Worms (Annelids)
 - Roundworms (Nematoda)
 - Flatworms (Platyhelminthes) and others.
- "Worm" describes any bilaterally symmetrical, legless, soft bodied animal at least 2-3 times as long as it is wide.

❖ Phylum Platyhelminthes

- Platyhelminthes
- (Platy= Flat; Heminthes =Worm):
 - Are dorsoventrally flattened.
 - Have evolved cephalization with their sense organs concentrated at the head.
 - Have a ladder-type nervous system.
 - Are bilaterally symmetrical.
 - triploblastic,
 - Acoelomate
 - Have evolved organs and in some cases organ systems.
 - The first (and simplest) excretory or osmoregulatory systems and circulatory systems are found in members of these groups.



- Platyhelminthes usually slender and leaflike or ribbonlike.
- Unsegmented.
- Have no circulatory or respiratory organs. Flat body increases surface area and allows the animal to exchange gas and lose wastes by diffusion.
- Freelifing and Parasitic forms



❖ Nutrition

- The digestive system includes a mouth, pharynx, and blind intestine (the gut is incomplete).
- Food is sucked into the intestine where a combination of extracellular and intracellular digestion takes place.
- Undigested food exits via the pharynx.
- In some groups, the digestive tract is absent and all nutrients are absorbed across the tegument



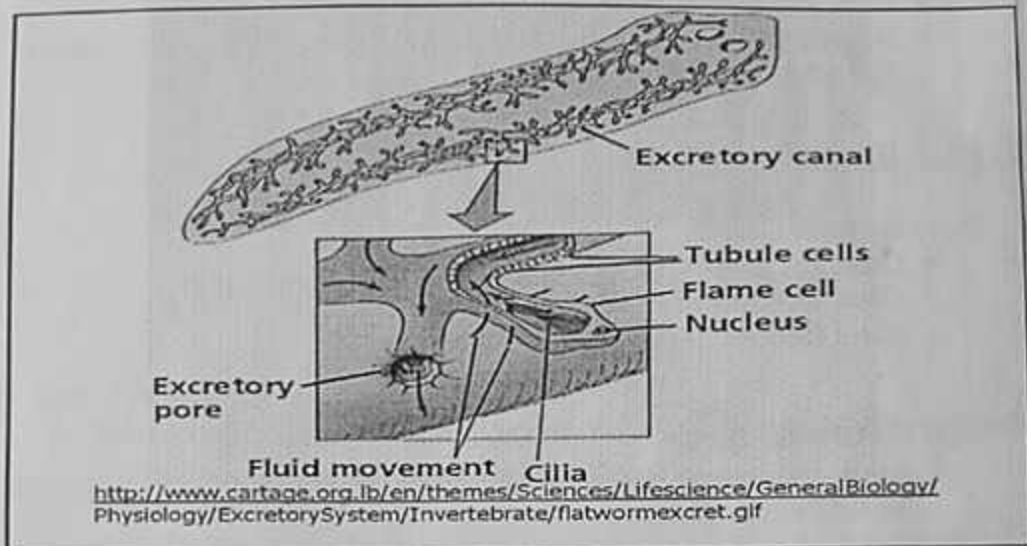
"Organization of the blind digestive cavity of polyclads [a group of Turbellarians] with highly branched diverticles (ventral view)"
<http://www.rzuser.uni-heidelberg.de/~bu6/Introduction04.html>

- Undigested food exits via the pharynx.
- In the Cestoda the digestive tract is absent and all nutrients are absorbed across the tegument (the syncytial membrane/body covering found in all parasitic Platyhelminthes).

❖ Excretion/Osmoregulation

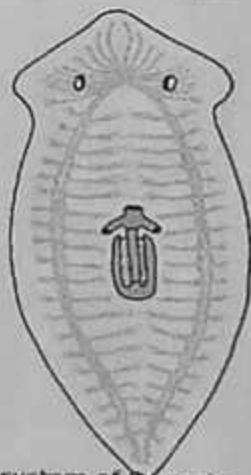
- The osmoregulatory system consists of a series of canals that end in flame cells or protonephridia.

- The flame cell consists of a fine meshed cup that contains cilia. The beating of the cilia draws fluid which is filtered as it passes into the cup.



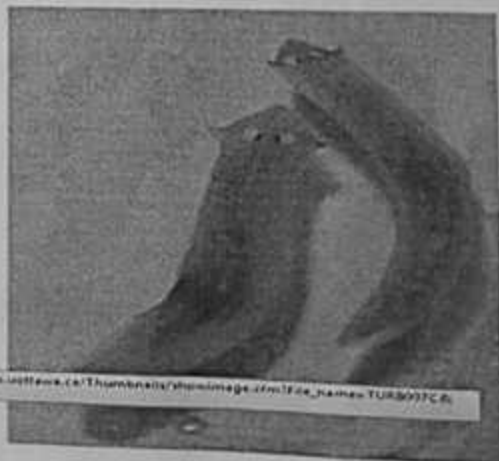
❖ Nervous system and sense organs

- A simple brain
- One to five pairs of longitudinal nerve cords
- In freshwater Planarians concentrations of sensory cells form two ear-like structures (the auricles) found on the side of the head.
- Light sensitive eyespots or ocelli are common in all classes but Cestoda.



Nervous system of Dugesia

<http://biodidac.bio.uottawa.ca/ftp/BIODIDAC/ZOO/PLATYHEL/DIAGCL/TURB007C.GIF>



❖ Reproduction

- Reproduction in the Platyhelminthes can be asexual or sexual.
- However, most are hermaphroditic and cross fertilize.
- In parasitic forms sexual and asexual reproduction may alternate in different stages of the life history

❖ Classification of Platyhelminthes

There are three classes:

- Class Turbellaria: free - living flatworms.
- Class Trematoda: endoparasitic flukes
- Class Cestoda: tapeworms

❖ Class Turbellaria

- Most species are marine and benthic (move around on the bottom in aquatic environments).
- Some also found in fresh water or in moist temperate and tropical terrestrial habitats.
- Free-Living (example *Dugesia tigrina*)

Figure 14.10

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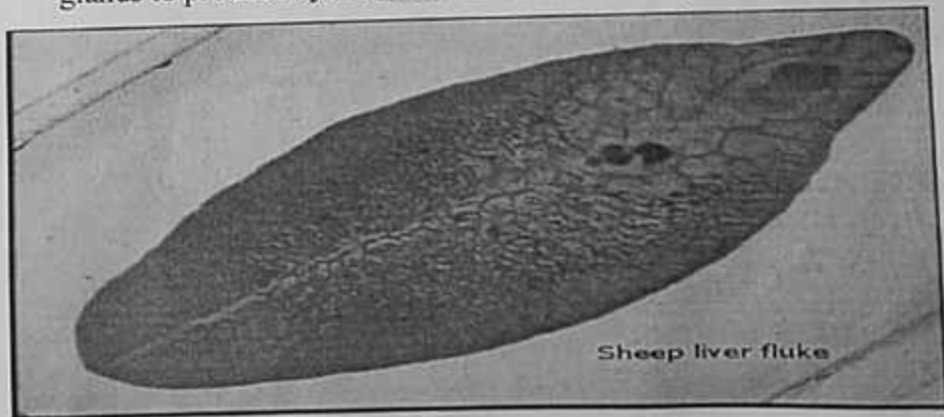
Marine turbellarian



Dugesia tigrina, a freshwater turbellarian

❖ Class Turbellaria

- Most Turbellarians are predators of invertebrates smaller than themselves.
- A few are herbivores or scavengers.
- In many species the pharynx is protrusible and can be inserted into the prey to begin digesting it.
- Some species stab prey with their penis (it has a hardened tip or stylet).
- Turbellarians move by swimming, creeping or crawling. They combine muscular contractions with ciliary movement to move.
- Turbellarians may also use waves of muscle movement to move.
- There are about 9000 species of trematodes (flukes) all of which are parasitic.
- Most parasitize vertebrates.
- Adaptations for parasitism include suckers and hooks for attachment, glands to produce cyst material and increased reproductive capacity.



Sheep liver fluke

❖ Class Trematoda

- However a major difference between turbellarians and trematode is the tegument.
- The tegument protects the parasite against its host (e.g. against digestive enzymes).

❖ SUBCLASSES

- Three major subclass:
 - Aspidogastrea
 - Endoparasites of molluscs, turtles and fishes.
 - Monogenea
 - Ectoparasites with a monoxenous life cycle in fishes.
 - Digenea
 - Endoparasites with a heteroxenous life cycle.

❖ Digenean Trematodes

- There are three subclasses of Trematodes, but two are small, poorly studied groups.
- The third group, the Digenea, however is a large group of major medical and economic importance.
- Predominantly endoparasites.
- Commonly referred to as flukes.
- Have varying predilection site; intestinal, liver, lung, blood.
- They cause important parasitic diseases of man and animals.
- Most successful.
- Three families
 - Fasciolidae.
 - Opisthochiidae.
 - Schistosomatidae.

❖ Life cycle of a typical trematode

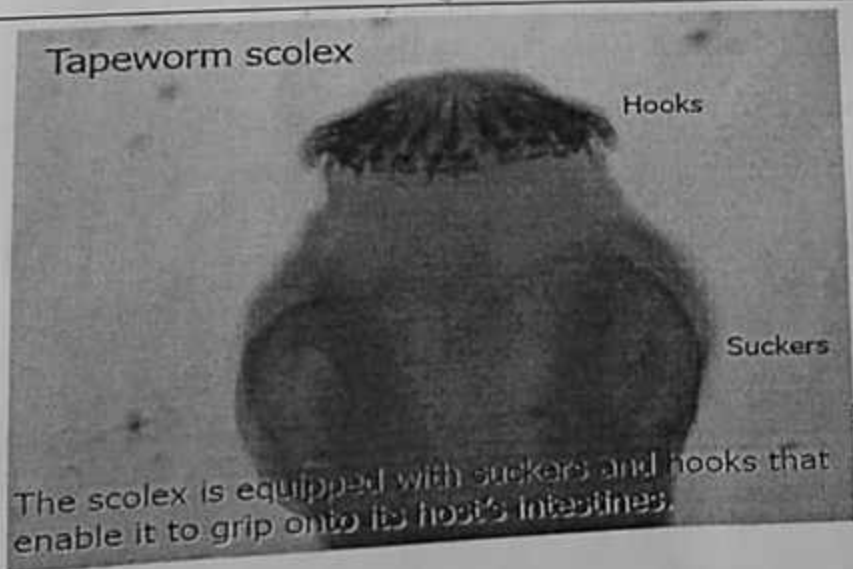
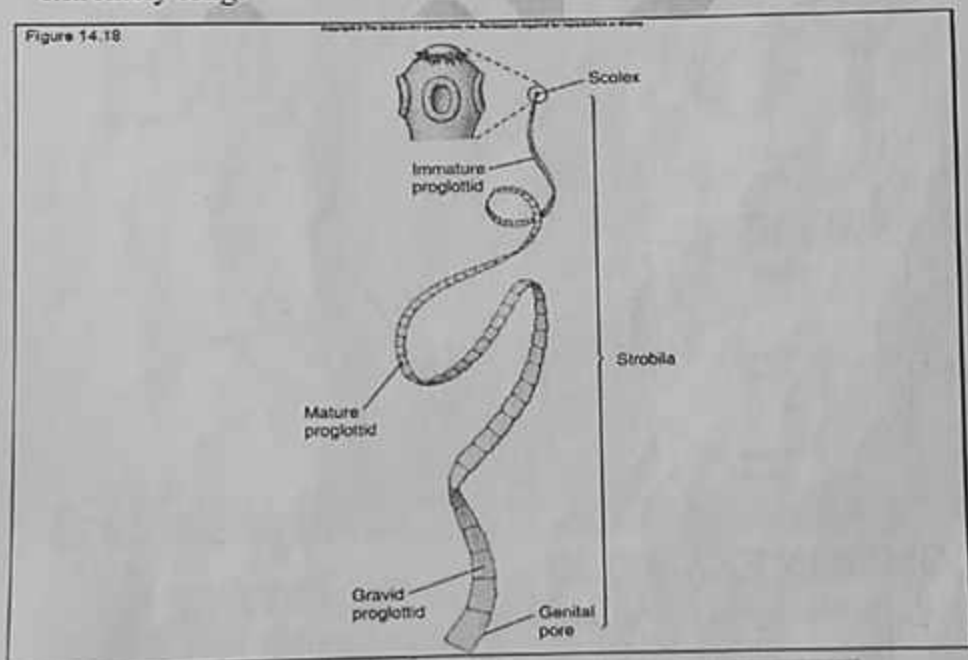
- Adult.
- Egg - shed in water.
- Miracidium (first larva stage) - penetrates a snail.
- Sporocyst.
- Redia.
- Cercaria - swim and penetrates a host.
- Metacercariae (Not applicable to all).

❖ Common examples

- *Schistosoma* spp. - Schistosomiasis (Blood fluke).
- *Fasciola* spp. - Fascioliasis (Liver fluke).
- *Paragonimus* spp. - Paragonimiasis (Lung fluke).

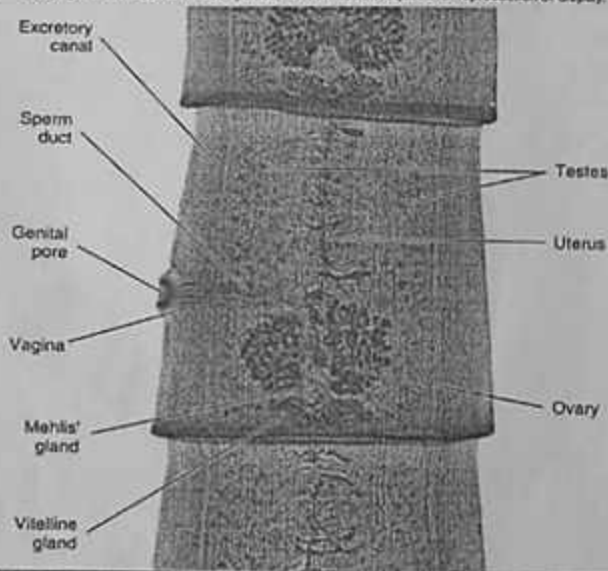
➤ Class Cestoda (tapeworms)

- Tapeworms are parasites of the vertebrate digestive tract and about 4000 species are known.
- Almost all tapeworms require at least two hosts with the definitive host being a vertebrate, although intermediate hosts can be invertebrates.
- They have long, flat, tape like bodies composed of a scolex for attaching to their host.
- A chain of many reproductive units or proglottids called strobila.
- New proglottids form behind the scolex and the strobila may become extremely long.



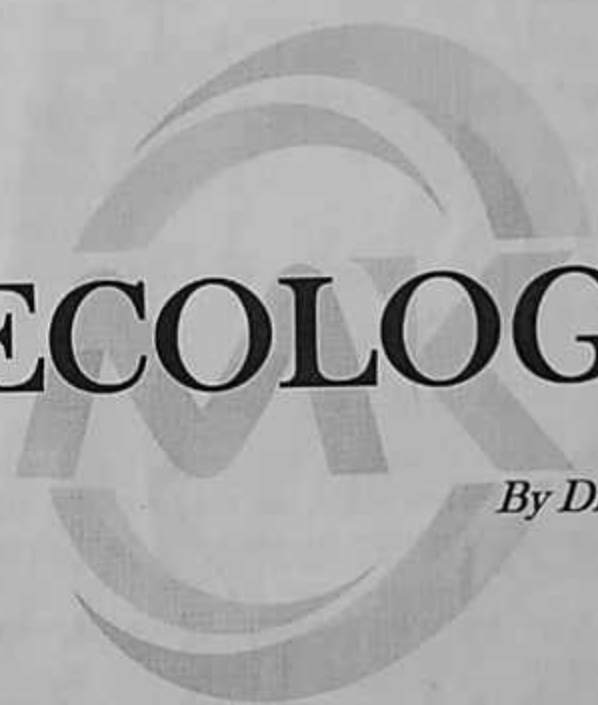
- Tapeworms live in the intestines and because they are immerse in digested food lack a digestive system of their own. Instead they simply absorb food across their tegument.
- To facilitate the absorption of food a tapeworm's tegument has huge numbers of tiny projections called **microtriches** which are broadly similar to the microvilli of the vertebrate intestine.
- They similarly increase the surface area of the tegument for absorption.
- Tapeworms are usually **monoecious** (have both male and female reproductive organs).
- A proglottid is fertilized by another proglottid in the same or a different strobila.
- Shell-encased embryos form in the uterus and exit the proglottid via a uterine pore or the entire proglottid may detach and pass out of the host.

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❖ COMMON EXAMPLES

- *Taenia saginata*—Beef tapeworm.
- *Taenia solium*—Pork tapeworm.
- *Diphyllobothrium latum*—Fish tapeworm.



ECOLOGY

By Dr OKOH H.I.

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❖ WHAT IS ECOLOGY?

- Ecology is the branch of biology that studies the relationships between living organisms, including humans, and their physical environment
- it seeks to understand the vital connections between plants and animals and the world around them
- Ecology also provides information about the benefits of ecosystems and how we can use Earth's resources in ways that leave the environment healthy for future generations

❖ What does ecology have to do with me?

- Protection of endangered species
- Biological control
- Afforestation
- Insecticidal plants (e.g. Neem plant)
- Natural products (E.g. Honey)

❖ Ecological Terminologies

- **Biosphere:** Is the part of the earth and the atmosphere where life can exist. This ranges from deep water bodies, land and few meters in the atmosphere
- **Species:** Is a group of organism that can interbreed to give rise to viable off springs
- **Population:** A group of organisms of the same species occupying the same area at a given time
- **Habitat:** Is a place where an organism lives e.g. land habitat is known as terrestrial and water habitat is called aquatic
- **Ecosystem:** is the basic functional unit capable of sustaining its self and consisting of plants and animals interacting with each other and non living
- **Community:** Is a collection of population of different organism occupying the same area or a group of different species of organism in the same area
- **Ecology niche:** This is a form used in relation to a particular organism. It refers to the role/profession of an organism in a given habitat e.g. the niche of spirogyra in a pond is to photosynthesis /to produce food
- **Biome:** These are very large ecological divisions found all over the world they have characteristic plant and animal communities e.g. equatorial rainforests biome, Amazon, Savannah woodland and grasslands, temperate forests etc.

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❖ BIODIVERSITY

- Short for biological diversity, biodiversity is the range of variation found among microorganisms, plants, fungi, and animals
- Some of this variation is found within species, such as differences in shapes and colours of the flowers of a single species of plants
- Biodiversity also includes the richness of species of living organisms on earth.

❖ Environment

The environment is the surroundings of an organism including the physical and chemical environment, and other organisms with which it comes into contact. This term is most frequently used in a human context, often referring to factors affecting our quality of life

❖ Natural Resources

Natural resources are living and nonliving materials in the environment that are used by humans

- There are two types: renewable (wildlife, fish, timber, water) and non-renewable (fossil fuels and minerals).
- Abiotic – Nonliving factors, which is a nonliving physical and chemical attribute of a system, for example light, temperature, wind patterns, rocks, soil, pH, pressure
- Biotic – of, pertaining to, or produced by life or living organisms (of an ecosystem)
- Adaptation – changes in an organism's physiological structure or function or habits that allow it to survive in new surroundings. For example, forests develop only where soil types, moisture and sunlight are balanced to the proper degree. Desert plants have made adjustments so as to be able to live under intense sunlight, on poor quality soils, and with reduced water supply
- Carnivore – a meat eater
- Commensalism – a symbiotic relationship between two species in which one species benefits and the other either benefits or harms. Often, the host species provides a home and/or transportation for the other species. Commensalism is much more difficult to demonstrate than mutualism. For true commensalism, the second species must be unaffected by the presence of the first, but commonly a detailed study of the relationship will show some affect on the second species

- Competition – when two or more organisms have the potential for using the same resource. May be Inter - or intra
- Saprophyte – an organism living on dead organic matter; neither parasitic nor making its own food
- Symbiosis – A close, prolonged association between two or more different organisms of different species that may, but does not necessarily, benefit each member



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